Schwarz Coupling for Solid Dynamics in Norma: Transmission Conditions, Interface Energy, and Energy-Consistent Design

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Abstract

This note is the consolidated reference for the Schwarz coupling conditions implemented in Norma and for the study of their energy behavior in dynamic solid mechanics. On a shared (nonoverlap) interface, the Robin condition is fully reflecting and the impedance condition is absorbing, with the impedance fixed by material data. On an overlap boundary, the impedance condition is consistent with the monodomain solution only when it carries the partner traction; with the consistent nodal traction the coupling is exact on conforming meshes (energy error 59% $\rightarrow$ 0.08% over 1 ms, with no adjustable parameters). On nonconforming meshes the energy error of the overlap coupling admits no a priori bound in sign or magnitude: because the two sides transfer partner data through independent, non-adjoint operators, the interface work is sign-indefinite — measured drifts vary non-monotonically from −53% to +174% to −66% across adjacent mesh ratios — and at a ten-millisecond horizon 74 of 108 configurations fail by element inversion after exponential energy growth. Every remedy attempted at the interface — variational transfer, exact traction transfer, filtered dissipation — fails for a structural reason: the persistent interface jump originates in the discrepancy between the two grids' discrete dispersion relations, not a transfer defect. Energy consistency requires a single-valued interface traction acting through L^2-adjoint transfer operators, which makes the interface power telescope to a sign-definite dissipation on the interface jump; this construction is available only on the nonoverlap variant, whose two sides share one interface. The resulting adjoint-paired nonoverlap coupling — one shared cross-mass matrix, force transfers adjoint to the partner's kinematic transfers, a harmonic-mean pair impedance, the dynamically consistent d'Alembert interface reaction, and an implicit–explicit treatment of the interface rows for explicit subdomains — never injects energy at any mesh ratio, conserves energy to 0.3% over 1 ms in all four integrator combinations on conforming and mildly nonconforming meshes, and tracks the monodomain trajectory over ten thousand steps. The note also records the subcycled (multi-time-step) exchange, the measured incompatibility of exact interface-work telescoping with a partitioned iteration, and the slot-keyed relaxation state required when controller stops exceed a subdomain time step. Practical recommendations and open questions conclude the note.

This document consolidates and supersedes three earlier notes: "Impedance-Matching vs. Robin–Robin Coupling Conditions in Schwarz Methods," "Variational Transfer for Impedance-Overlap Schwarz on Non-Conforming Meshes," and "Interface Energy Loss of the Impedance-Overlap Schwarz Coupling on Nonconforming Meshes, and an Energy-Consistent Redesign" (June–July 2026).

Contents

1 Setting

1.1 Decompositions and notation

Consider a solid body $\Omega \subset \mathbb{R}^3$ decomposed into two subdomains Ω_1 and Ω_2 that are integrated independently and coupled by a Schwarz iteration. The Schwarz alternating framework itself — time controller, subdomain solves, convergence criteria — follows Mota, Tezaur, and Phlipot [1] (the contact variant is described in [2]); this note is concerned with the transmission conditions imposed on the coupling boundaries and with their energy behavior. Norma supports two geometric configurations.

Nonoverlap. The subdomains partition Ω and share an interface $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ on which coupling is imposed. Let $\boldsymbol{n}_k$ be the outward unit normal to Ω_k on Γ (so $\boldsymbol{n}_1 = -\boldsymbol{n}_2$) and let $\boldsymbol{t}_k = \boldsymbol{\sigma}_k\,\boldsymbol{n}_k$ denote the traction on Ω_k's side. Continuity of the coupled solution requires

$$\boldsymbol{u}_1 = \boldsymbol{u}_2 \qquad \text{(kinematic continuity)}, \tag{1}$$

$$\boldsymbol{t}_1 + \boldsymbol{t}_2 = \boldsymbol{0} \qquad \text{(traction balance)}. \tag{2}$$

Classical Dirichlet–Neumann Schwarz imposes (1) on one side and (2) on the other. The Robin and impedance variants studied here replace both with a linear combination of displacement, velocity, and traction (Sec. 2).

Overlap. The subdomains properly intersect: $\Omega_1 \cap \Omega_2 \neq \emptyset$. Each Ω_k has a portion of its boundary, $\Gamma_k \subset \Omega_j$ with $j \neq k$, lying strictly in the interior of the partner. That boundary is not shared with Ω_j, so no mutual traction is directly available there; on the other hand, the partner provides a well-defined interior state at every point of Γ_k. The coupling conditions available on an overlap boundary are those that can be built from that interior state (Sec. 3).

Throughout, $\boldsymbol{W}_k = \int_{\Gamma_k} \boldsymbol{N}_k\,\boldsymbol{N}_k^\top \,\mathrm{d}\Gamma$ denotes the boundary mass matrix of side k, $\boldsymbol{P}$ a kinematic transfer operator mapping partner data to a side's interface nodes, a subscript j the partner of side k, and $[\![\cdot]\!]$ an interface jump. All interface operators ($\boldsymbol{W}_k$, the transfer operators, and the cross-mass matrix of Sec. 9) are assembled once at initialization, on the reference configuration, with the standard facet Gauss rule of the receiving side set; they are not updated with the deformation.

1.2 Benchmark and energy metrics

All measurements in this note come from one benchmark: a bent 3D cantilever released from a parabolic initial displacement ($E = 6.895\,\mathrm{GPa}$, $\nu = 0.25$, $\rho = 2768\,\mathrm{kg/m^3}$, bar length $0.254\,\mathrm{m}$, $\Delta t = 10^{-6}\,\mathrm{s}$ unless stated, horizons of $1\,\mathrm{ms}$ and $10\,\mathrm{ms}$, Schwarz iterated to convergence at relative tolerance 10^{-8} at every step). Subdomain integrators are trapezoidal Newmark (implicit, I) and central difference (explicit, E), in the four pairings II, IE, EI, EE (first letter: clamped subdomain). Overlap decompositions split the bar lengthwise with overlap widths $\{0.0508, 0.0254, 0.0127\}\,\mathrm{m}$; the nonoverlap decomposition splits at $x = 0.127\,\mathrm{m}$. Nonconforming pairs fix the free subdomain's mesh at $h = 6.35\,\mathrm{mm}$ and coarsen the clamped subdomain to $6.35/r\,\mathrm{mm}$, written as ratio 1:r;

the original three-point family used $r \in \{1.0,\ 0.75,\ 0.5\}$, i.e. clamped-side meshes of 6.35, 8.47, and 12.7 mm. (The superseded working notes labeled the ratios 1:0.75 and 1:0.5 as 3:4 and 1:2, quoting h_{free}:h_{clamped} directly; this note uses the 1:r convention throughout, as do the archived data directories' drift tables.)

Time-controller vocabulary. The multi-domain time controller advances in *stops*. Within a stop, each subdomain advances to the stop end in its own *substeps*, and the Schwarz iteration performs *sweeps* — one solve of every subdomain per sweep, exchanging interface data between sweeps — until the exchanged data converge, at which point the stop is committed. The exchanged datum may be under-relaxed between sweeps with a relaxation parameter $\theta \in (0, 1]$ ($\theta = 1$ means no relaxation), either fixed or adapted by Aitken Δ^2 extrapolation from successive residuals. A stop spanning several substeps is called a *window* (Sec. 12). A mesh pair is *nested* when every coarse node coincides with a fine node.

Explicit stability margin. The explicit runs use central difference at the same $\Delta t = 10^{-6}$ s as the implicit runs. A monodomain time-step sweep on the finest ($h = 6.35$ mm) mesh brackets the central-difference stability limit between 3.5 and 3.7 μs — every run at $\Delta t \leq 3.5\,\mu$s conserves the staggered energy to machine precision over 2000 steps, and a run at 3.7 μs fails within 14 steps — consistent with the one-dimensional axial-mode bound $h/c_p = 3.67\,\mu$s, which the lumped-mass hexahedral mesh attains. Every explicit subdomain in this study therefore operates at or below 0.27 of its stability limit (coarser meshes have proportionally larger limits; the subcycled explicit runs operate at 0.07), so no result reported here is affected by proximity to the explicit stability limit. The `CFL` input parameter of the central-difference integrator is a per-step clamp multiplier on an element-diameter estimate recomputed on the current configuration; for the cube elements used here that estimate is $\sqrt{3}\,h/c_p \approx 6.4\,\mu$s, about 1.7× the measured limit, so the clamp at the setting `CFL: 1.0` never engaged in these runs and would not by itself guarantee stability — a run clamped to 6.36 μs fails within two steps. With the diameter-based estimate, clamp settings of about 0.5 or below are required for the clamp to be protective on hexahedral meshes.

Partition-of-unity (PU) energy. Summing the two subdomains' energies counts the overlap region twice and distorts the diagnostic: a provably conservative run shows apparent ±13% excursions as the pulse transits the overlap. All energies quoted here count the overlap once, through a partition of unity. The production implementation (`blended energy output: true`; `src/overlap_energy.jl`) uses a smooth Arlequin distance-ratio partition; the study's post-processor weights doubly-covered elements by 1/2 and reproduces Norma's unweighted output to at least four digits. Drift is reported as $E(t)/E(0) - 1$ in percent, with $E(0) \approx 1506$ J.

Staggered energy for central difference. Central difference advances the lumped-mass system with leapfrog structure, so the discrete energy it conserves exactly on linear problems is the staggered lumped-mass form

$$E_n \;=\; \tfrac{1}{2}\,\dot{\boldsymbol{u}}_{n-\frac{1}{2}}^T \boldsymbol{M}_L\, \dot{\boldsymbol{u}}_{n+\frac{1}{2}} + \mathrm{SE}(\boldsymbol{u}_n) \;=\; \tfrac{1}{2}\,\dot{\boldsymbol{u}}_n^T \boldsymbol{M}_L\, \dot{\boldsymbol{u}}_n - \tfrac{\Delta t^2}{8}\,\boldsymbol{a}_n^T \boldsymbol{M}_L\, \boldsymbol{a}_n + \mathrm{SE}(\boldsymbol{u}_n), \tag{3}$$

where $\boldsymbol{M}_L$ is the lumped mass and SE the stored energy. Evaluating instead the consistent-mass kinetic energy at integer steps misrepresents the short-wavelength content of the sharp release front in two ways: the consistent and lumped Rayleigh quotients diverge at high wavenumber, and the integer-step velocity aliases the staggered content. A monodomain central-difference run of this benchmark reads −8.2% at 1 ms in the consistent-mass integer-step metric and ±0.000% in (3).

Norma's energy output evaluates (3) for central-difference subdomains, and explicit-subdomain results below use this metric unless noted.

Organization. Sections 2 and 3 define the four transmission conditions. Section 4 establishes the conforming baseline. Sections 5 and 6 record the nonconforming overlap problem: first the transfer-fidelity study and its conclusion, then the parametric energy study and its diagnosis. Section 7 states the theoretical requirements for energy consistency and Sec. 8 their consequences for the overlap family. Sections 9–12 present the adjoint-paired nonoverlap coupling, subcycling, long-horizon behavior, and the windowed controller stops. Section 13 collects findings, recommendations, and open questions.

2 Transmission conditions on a shared interface

2.1 Robin coupling

On each side of the interface a Robin parameter $\alpha > 0$ is prescribed in the input and the condition

$$\boldsymbol{t}_k + \alpha\,\boldsymbol{u}_k \;=\; \boldsymbol{g}_k \qquad \text{on } \Gamma, \quad k = 1, 2, \tag{4}$$

is imposed, with the datum built from the partner's previous Schwarz iterate, $\boldsymbol{g}_k^{(n+1)} = -\boldsymbol{t}_j^{(n)} + \alpha\,\boldsymbol{u}_j^{(n)}$, $j \neq k$. At the Schwarz fixed point $\boldsymbol{u}_k = \boldsymbol{u}_j$ and $\boldsymbol{t}_k = -\boldsymbol{t}_j$, so (4) reproduces (1) and (2) simultaneously.

Discretely, the self term contributes the boundary stiffness $\boldsymbol{K}_{\mathrm{rs}} = \alpha\,\boldsymbol{W}$ to the tangent, and the partner datum enters the boundary force as $\boldsymbol{f}_{\Gamma,k} \leftarrow \boldsymbol{f}_{\Gamma,k} - \boldsymbol{t}_j\big|_\Gamma + \alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j$. The parameter α has units of stiffness per unit area (N/m^3) and interpolates between a Dirichlet constraint ($\alpha \to \infty$) and a Neumann condition ($\alpha \to 0$); the optimal value minimizes the contraction factor of the Schwarz iteration (fastest convergence), and in practice the scaling $\alpha \sim E/h$ has been effective.

Invocation and guards. In Norma the Robin condition is the input keyword `Schwarz RR nonoverlap`. Since 2026-08-25 this keyword is no longer an alias for the impedance condition: it forces $Z = 0$, requires a positive `robin parameter` (the Robin spring is the condition's only coupling term), rejects the `impedance scale` key with a hard error pointing at `Schwarz impedance nonoverlap`, and defaults to `adjoint pairing: false`, so per-side values of α are allowed as in the classical Robin–Robin literature. `adjoint pairing: true` remains available and makes the Robin spring a conservative paired interface spring with one shared α (Sec. 9). Because the condition is fully reflecting (Sec. 2.3) and can pump energy at a dynamic interface (observed as a growing interface mode under pure torsion; Norma issue #176), a warning is issued whenever it is combined with a dynamic time integrator; the keyword is intended for quasi-statics and for comparison studies. Both nonoverlap keywords are implemented by the same boundary-condition type — the Robin condition is the $Z = 0$ member of the impedance family — and differ only in their defaults and input guards, so the degenerate exchange with neither dashpot nor Robin spring is unreachable from either keyword.

2.2 Impedance coupling

For a semi-infinite elastic bar with density ρ and P-wave modulus $M = \lambda + 2\mu$, a purely outgoing wave $u(x,t) = f(x - c_p t)$ with $c_p = \sqrt{M/\rho}$ satisfies $\dot{u} = -c_p\,u_x$, and with $\sigma = M\,u_x$,

$$\sigma + Z\,\dot{u} \;=\; 0, \qquad Z \;=\; \rho\,c_p \;=\; \sqrt{\rho\,(\lambda + 2\mu)}, \tag{5}$$

the Lysmer–Kuhlemeyer nonreflecting boundary condition [3], with Z the characteristic acoustic impedance. The impedance Schwarz condition augments (4) with this absorbing term:

$$\boldsymbol{t}_k + Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k \;=\; \boldsymbol{g}_k \qquad \text{on } \Gamma, \qquad \boldsymbol{g}_k^{(n+1)} \;=\; -\boldsymbol{t}_j^{(n)} + Z\,\dot{\boldsymbol{u}}_j^{(n)} + \alpha\,\boldsymbol{u}_j^{(n)}. \tag{6}$$

At the fixed point the impedance and displacement terms cancel and (6) again reproduces continuity and traction balance; the impedance is computed from material data and requires no input, while α remains an optional input parameter. Concretely, each side's estimate is $Z_k = \sqrt{\rho_k(\lambda_k + 2\mu_k)}$ from its own subdomain's material constants (the implementation reads the material assigned to the first entry of the subdomain's block–material mapping, so each subdomain is treated as acoustically homogeneous for this estimate), multiplied by the input `impedance scale` (default 1; required positive — a zero scale would degenerate to the Robin condition, which has its own keyword). The input keyword is `Schwarz impedance nonoverlap`, with `adjoint pairing: true` as its default (Sec. 9).

Under Newmark integration with parameters β, γ the new-step velocity is linear in the new-step displacement with coefficient $c_v = \gamma/(\beta\,\Delta t)$, so the self term contributes the tangent $\boldsymbol{K}_{\text{is}} = (Z\,c_v + \alpha)\,\boldsymbol{W}$ and the self force $\boldsymbol{f}_{\text{is}} = \boldsymbol{W}(Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k)$; the partner datum enters the boundary force through the same surface operator and transfer as the Robin case, with the additional velocity term. In the quasi-static limit Norma sets $c_v = 0$ and the impedance condition reduces to the Robin condition (4); note that a quasi-static impedance run with $\alpha = 0$ degenerates to a pure Neumann condition and does not couple the subdomains kinematically.

2.3 Reflection at the interface

The essential difference between the two conditions is their treatment of outgoing waves. Place the interface at $x = 0$ with Ω_k in $x < 0$ and consider a harmonic incident wave of amplitude F and a reflected wave of amplitude R. Imposing the Robin condition $\sigma + \alpha u = 0$ yields the reflection coefficient

$$R_{\text{RR}}(\omega) \;=\; \frac{\mathrm{i}\,\omega\,Z - \alpha}{\mathrm{i}\,\omega\,Z + \alpha}, \qquad |R_{\text{RR}}(\omega)| \equiv 1, \tag{7}$$

so that for any real $\alpha > 0$ the elastic interface fully reflects the incoming wave with a frequency-dependent phase shift, and no wave energy crosses the interface. Imposing the impedance condition $\sigma + Z\dot{u} + \alpha u = 0$ yields

$$R_{\text{imp}}(\omega) \;=\; -\frac{\alpha}{2\,\mathrm{i}\,\omega\,Z + \alpha}, \qquad |R_{\text{imp}}(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + 4\,\omega^2 Z^2}} \;<\; 1, \tag{8}$$

with $R_{\text{imp}} \equiv 0$ at $\alpha = 0$: outgoing waves are absorbed into the partner at all frequencies. This removes the reflection mechanism by which Robin coupling can accumulate energy in dynamic problems, particularly when the two subdomains use different time integrators.

3 Transmission conditions on an overlap boundary

3.1 Dirichlet coupling

The classical overlap Schwarz condition replaces the free boundary of Ω_k on Γ_k with an imposed kinematic trace drawn from the partner,

$$\boldsymbol{u}_k = \boldsymbol{u}_j, \qquad \dot{\boldsymbol{u}}_k = \dot{\boldsymbol{u}}_j, \qquad \ddot{\boldsymbol{u}}_k = \ddot{\boldsymbol{u}}_j \qquad \text{on } \Gamma_k, \tag{9}$$

constructed in one of two ways. *Pointwise interpolation*: for each overlap node, locate the partner element containing it in the reference configuration, evaluate the partner shape functions there, and assign the interpolated value; the corresponding degrees of freedom are then constrained. *Weak (L^2) projection*: precompute the projector $\boldsymbol{P}_{kj} = \boldsymbol{W}_k^{-1}\boldsymbol{L}_{kj}$, where $\boldsymbol{L}_{kj} = \int_{\Gamma_k} \boldsymbol{N}_k\,(\boldsymbol{N}_j^{\mathrm{vol}})^\top\,\mathrm{d}\Gamma$ is the cross-mass matrix coupling the partner's volume shape functions to the overlap boundary, and assign $\boldsymbol{u}_k = \boldsymbol{P}_{kj}\,\boldsymbol{u}_j$. Both variants are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` (`use_weak` flag). Because (9) constrains the overlap degrees of freedom, Dirichlet coupling adds no interface tangent term.

3.2 Impedance coupling

The impedance overlap condition leaves the overlap degrees of freedom free and couples Ω_k to the partner through a full zeroth-order optimized-Schwarz transmission condition,

$$\boldsymbol{t}_k - \boldsymbol{t}_j + \boldsymbol{Z}\big(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j\big) + \alpha\big(\boldsymbol{u}_k - \boldsymbol{u}_j\big) = \boldsymbol{0} \qquad \text{on } \Gamma_k, \tag{10}$$

where $\boldsymbol{t}_j = \boldsymbol{\sigma}_j\,\boldsymbol{n}_k$ is the partner's traction evaluated on Γ_k with Ω_k's outward normal, and the impedance is the P/S-split tensor

$$\boldsymbol{Z} = Z_p\,\boldsymbol{n}_k \otimes \boldsymbol{n}_k + Z_s\big(\boldsymbol{I} - \boldsymbol{n}_k \otimes \boldsymbol{n}_k\big), \qquad Z_p = \sqrt{\rho(\lambda + 2\mu)}, \quad Z_s = \sqrt{\rho\mu}, \tag{11}$$

with ρ, λ, μ taken from the partner's material (optimized-Schwarz cross-scaling: each side's optimal transmission operator approximates the neighbor's Dirichlet-to-Neumann map, so at a bimaterial interface each subdomain carries the other's characteristic impedances).

The partner traction is essential for consistency. The true monodomain solution has continuous kinematics and a generally nonzero traction $\boldsymbol{t}$ on the artificial surface Γ_k; it satisfies (10) identically. Dropping $-\boldsymbol{t}_j$ (as an early implementation did) turns the condition into a relative dashpot, $\boldsymbol{t}_k = \boldsymbol{Z}(\dot{\boldsymbol{u}}_j - \dot{\boldsymbol{u}}_k) + \cdots$, which the monodomain solution fails by exactly $\boldsymbol{t}$: the converged coupled solution then carries a spurious interface velocity jump $\approx \boldsymbol{t}/Z$ whose dissipated power $\sim |\boldsymbol{t}|^2/Z$ acts as permanent interface damping (59% energy loss over 1 ms on the conforming benchmark; Sec. 4).

Evaluating the partner traction. Because Γ_k is an interior surface of the partner mesh, the partner's assembled internal force cannot supply $\boldsymbol{t}_j$ there (interior rows carry the inertia residual, not $\boldsymbol{\sigma}\,\boldsymbol{n}$). Norma evaluates $\boldsymbol{t}_j$ in one of two ways, selected by the `partner traction` option (default `auto`).

Consistent traction (node-aligned interfaces). When every node of Γ_k coincides with a partner node (detected automatically from the interpolation weights), the weak partner traction is extracted as the consistent nodal reaction: for each interface node i,

$$\big[\boldsymbol{W}\,\boldsymbol{t}_j\big]_i = \int_{\Gamma_k} \boldsymbol{t}_j\,\varphi_i\,\mathrm{d}\Gamma = -\sum_{e\,\in\,\mathcal{E}_i^{\mathrm{ext}}} \big(\boldsymbol{f}_e^{\mathrm{int}} + \boldsymbol{M}_e\,\ddot{\boldsymbol{u}}_e\big)_i, \tag{12}$$

where $\mathcal{E}_i^{\mathrm{ext}}$ is the single layer of partner elements adjacent to Γ_k on its exterior side — precisely the elements Ω_k is missing from the monodomain assembly — and $\boldsymbol{M}_e$ is the partner's element mass (consistent or lumped, matching the partner's integrator). With this definition the monodomain solution satisfies Ω_k's coupled discrete equations *identically*: the transmission condition transmits all discrete content, including the mesh-scale part that nodal stress recovery misrepresents. Implemented by `build_consistent_traction_patch!` and `consistent_partner_traction` (`schwarz.jl`).

Recovered-stress traction (fallback). On nonconforming interfaces $\boldsymbol{t}_j$ is obtained by interpolating the partner's nodal-recovered stress with the same shape-function data used for the kinematic fields and contracting with Ω_k's outward normals. Consistent (L^2-projected, Cholesky-factorized) nodal recovery is enabled automatically for any model carrying this condition; lumped recovery's $O(h)$ nodal-stress error enters the transmission condition directly and dominates the residual energy error (13.6% vs. 1.5% with consistent recovery; Sec. 4).

Discrete assembly. The self tangent consists of 3×3 nodal blocks $\boldsymbol{K}_{\mathrm{is},ij} = W_{ij}\,(c_v\,\boldsymbol{Z}(\boldsymbol{n}_j)+\alpha\,\boldsymbol{I})$; the self force is $\boldsymbol{f}_{\mathrm{is}} = \boldsymbol{W}(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_k+\alpha\,\boldsymbol{u}_k)$ with $\boldsymbol{Z}$ applied nodally before the surface weighting; and the partner datum enters as $\boldsymbol{f}_\Gamma \leftarrow \boldsymbol{f}_\Gamma + \boldsymbol{W}\,\boldsymbol{t}_j + \boldsymbol{W}(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_j+\alpha\,\boldsymbol{u}_j)$, with the kinematic fields transferred variationally by default (Sec. 8) or by pointwise interpolation as a legacy option. At the Schwarz fixed point all three jumps in (10) vanish for the monodomain solution, so on conforming meshes the converged coupled solution is the monodomain solution and the condition needs no parameter tuning: $\boldsymbol{Z}$ is fixed by the partner material and α may be zero. A step-dependent multiplier on $\boldsymbol{Z}$ (`impedance_scale`) rescales the whole operator; with the traction term in place the conserving scale is the physical one (scale = 1), and the historical role of the scale as a per-problem tuning parameter — which compensated the missing-traction dissipation — is obsolete.

No separate Robin overlap variant. A Robin condition requires a partner traction for its datum; on an overlap boundary that traction is available only through the machinery above, and the resulting condition is exactly the $\boldsymbol{Z}=\boldsymbol{0}$ special case of (10). It is subsumed by the impedance overlap condition and not exposed separately.

3.3 Summary of the four conditions

Tables 1 and 2 collect the four conditions and their discrete ingredients.

Table 1: Transmission conditions on a shared (nonoverlap) interface.

	Robin	Impedance
Boundary condition	$\boldsymbol{t}_k+\alpha\,\boldsymbol{u}_k=\boldsymbol{g}_k$	$\boldsymbol{t}_k+Z\,\dot{\boldsymbol{u}}_k+\alpha\,\boldsymbol{u}_k=\boldsymbol{g}_k$
Self tangent	$\alpha\,\boldsymbol{W}$	$(Z\,c_v+\alpha)\,\boldsymbol{W}$
Self internal force	$\alpha\,\boldsymbol{W}\,\boldsymbol{u}_k$	$\boldsymbol{W}\,\big(Z\,\dot{\boldsymbol{u}}_k+\alpha\,\boldsymbol{u}_k\big)$
Partner datum	$-\boldsymbol{t}_j+\alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j$	$-\boldsymbol{t}_j+Z\,\boldsymbol{W}\,\boldsymbol{P}\,\dot{\boldsymbol{u}}_j+\alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j$
Z set by	(not used)	material (automatic)
α set by	input file	input file, optional
Quasi-static limit	same	reduces to Robin
Reflection $\lvert R\rvert$	$\equiv 1$ (fully reflecting)	<1; 0 at $\alpha=0$

4 Conforming meshes: the partner-traction correction sequence

On the conforming, node-aligned overlap pair (II, PU metric, physical impedance, $\alpha=0$), the energy error of the impedance overlap coupling over 1 ms decreases through four successive corrections, all parameter-free:

$$\underbrace{59\%}_{\text{missing } \boldsymbol{t}_j} \longrightarrow \underbrace{14\%}_{+\,\text{traction term (lumped recovery)}} \longrightarrow \underbrace{1.5\%}_{+\,\text{consistent recovery}} \longrightarrow \underbrace{0.08\%}_{+\,\text{consistent traction}}$$

Table 2: Transmission conditions on an overlap boundary.

	Dirichlet	Impedance
Boundary condition	$\boldsymbol{u}_k = \boldsymbol{u}_j$ (pointwise or L^2)	$\boldsymbol{t}_k - \boldsymbol{t}_j + \boldsymbol{Z}\,[\![\dot{\boldsymbol{u}}]\!] + \alpha\,[\![\boldsymbol{u}]\!] = \boldsymbol{0}$
Self tangent	(DOFs constrained)	$W_{ij}\left(c_v\,\boldsymbol{Z}(\boldsymbol{n}_j) + \alpha\,\boldsymbol{I}\right)$
Self internal force	(none)	$\boldsymbol{W}\left(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k\right)$
Partner datum	$\boldsymbol{u}_k \leftarrow \boldsymbol{P}\,\boldsymbol{u}_j$	$\boldsymbol{W}\left(\boldsymbol{t}_j + \boldsymbol{Z}\,\dot{\boldsymbol{u}}_j + \alpha\,\boldsymbol{u}_j\right)$
Partner traction $\boldsymbol{t}_j$	(not used)	consistent (12) or recovered stress
$\boldsymbol{Z}$ set by	(not used)	partner material (automatic), P/S split
α set by	(not used)	input file, optional

against the 0.1% reference of converged conforming Dirichlet coupling, which the final step matches.

Diagnosis of the recovered-stress residual. The residual of the recovered-stress step (1.5% with a scalar impedance, 3.9–4.1% with the tensor impedance (11)) was isolated with an interface-work instrumentation (`report_impedance_interface_work`, environment-gated, CSV output via `NORMA_IMPEDANCE_WORK_CSV`) that decomposes the interface power into partner-traction, dashpot, and Robin channels; the dashpot and Robin powers vanish identically for the exact solution, so their integrated work measures spurious dissipation directly. At every converged stop the instrumentation also records the root-mean-square (RMS) velocity jump over the coupling-boundary nodes, $\left(\sum_i |\dot{\boldsymbol{u}}_j - \dot{\boldsymbol{u}}_k|_i^2/n\right)^{1/2}$ for the n nodes i of Γ_k; the velocity-jump values quoted in Tables 3 and 7 are averages of this per-stop RMS over the run. Three experiments discriminated the candidate mechanisms: (i) sweeping the Schwarz tolerance from 10^{-8} to 10^{-11} leaves the energy history bit-identical, ruling out iteration lag; (ii) the loss is invariant under $\Delta t \in \{2, 1, 0.5\} \times 10^{-6}$ s, ruling out the time discretization; (iii) the loss halves when h is halved (3.9% → 1.9%, tensor impedance): first order in mesh size. The measured velocity jump satisfies $\boldsymbol{Z}(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j) \approx \boldsymbol{t}_j - \boldsymbol{t}_k$ throughout — exactly what the converged transmission condition dictates — so the mechanism is: the recovered-stress traction error on mesh-scale field content is converted by the transmission condition into an interface velocity jump $\sim \delta t/Z$, which the dashpot dissipates at rate $|\delta t|^2/Z$ over the whole run. Replacing the recovered traction with the consistent traction (12) eliminates it: energy loss 3.93% → 0.076% and integrated dashpot work −42.6 J → +0.1 J on the same benchmark (tensor impedance). This also explains why Dirichlet overlap coupling is exact on conforming meshes — nodal injection transmits all discrete content with no recovery step.

P/S tensor split. The tensor impedance (11) is exact for body waves at normal incidence and is the correct operator for bimaterial interfaces. With the recovered-stress traction it appeared slightly detrimental on this homogeneous, flexure-dominated benchmark (conforming: 4.1% vs. 1.5% scalar); the diagnosis above resolves this: the tangential traction (recovery) error, which dominates in flexure, is dissipated at rate $|\delta t|^2/Z_s$, and $Z_s = Z_p/\sqrt{3}$ amplifies it. With the consistent traction the error term is gone and the tensor impedance conserves to 0.076%. Transmission conditions should be benchmarked on conforming meshes; nonconforming results can overstate a scheme's accuracy through error cancellation (Sec. 5).

Impedance scale. With the traction term missing, a swept constant multiplier on Z interpolates between over-absorbing and unstable behavior, and the best scale on the nonconforming pair was 1.12 — a cancellation of the missing-traction dissipation against nonconforming transfer injection, hence mesh- and problem-specific. With the traction term in place the optimum is the physical scale 1.0; historical per-simulation tuning of the scale (and of large α values near 5×10^{11}–10^{12}) optimized a double-counted diagnostic, not the physics.

5 Nonconforming overlap: transfer fidelity

With the conforming case resolved, the nonconforming overlap interface was studied on the 1:0.75 pair ($h = 8.47$ vs. $6.35\,\mathrm{mm}$). The net drift appears small ($\pm 1\%$ at 1 ms), but the channel decomposition (Table 3) shows this is a cancellation of two error channels, each transferring roughly half the system energy per millisecond.

Table 3: Interface-work channels of the impedance overlap coupling with pointwise transfer and recovered-stress traction (1 ms, II, PU metric, overlap width 0.0508 m, tensor impedance, mesh ratio 1:0.75 for the nonconforming columns). The small net drift on the nonconforming pair is the difference of two large channels. The conforming net column shows the recovered-stress → consistent-traction transition of Sec. 4; the velocity-jump row is the run average of the per-stop RMS jump defined there.

	conforming	nonconforming h	nonconforming $h/2$
dashpot (spurious) work	−42.6 J (2.8%)	−606.5 J (40%)	−598.5 J (40%)
traction-channel work	—	+677 J (45%)	+768 J (51%)
net PU-energy drift	3.9% → 0.08%	−0.75%	+1.5%
velocity jump (RMS)	8.9 m/s	40 m/s (growing 15 → 60)	27 m/s

Three facts motivated the investigation: (i) the channels do not shrink under mesh refinement, so the pointwise-interpolation transfer introduces an $O(1)$ error on mesh-scale content into the condition; (ii) the net error is the difference of two large channels and changes sign between h and $h/2$ — it is not robust; (iii) the velocity jumps grow through the run: each interface pass leaves residual mesh-scale content that is repeatedly re-transmitted between the grids.

5.1 The variational transfer and the fidelity sequence

Gander, Halpern, and Nataf [7] state three conditions for discrete energy convergence of waveform-relaxation coupling on nonmatching grids: (i) Robin (not Dirichlet) transmission; (ii) consistent boundary discretization of the transmission operator; (iii) an L^2-contractive intergrid transfer. The impedance overlap condition satisfies (i) and (ii); pointwise interpolation violates (iii), while the variational projector $\boldsymbol{P}_{kj} = \boldsymbol{W}_k^{-1}\boldsymbol{L}_{kj}$ (Sec. 3.1) is an $L^2(\Gamma_k)$-orthogonal projection, hence nonexpansive by construction, and transfers constants exactly for any quadrature rule (partition of unity gives $\boldsymbol{L}_{kj}\boldsymbol{1} = \boldsymbol{W}_k\boldsymbol{1}$).

A sequence of transfer improvements was implemented and measured on the instrumented 1 ms benchmark, each verified independently (the consistent constructions pass a uniform-stress patch test to 10^{-11} relative error):

- **Variational kinematic transfer** (`transfer: variational`): project $\dot{\boldsymbol{u}}_j$, $\boldsymbol{u}_j$, and the recovered stress variationally onto the receiving trace space instead of interpolating pointwise. On node-aligned interfaces the projector reduces to the identity, so the conforming limit is automatic.

- **Subdivided transfer quadrature** (`transfer quadrature subdivisions`): the integrand of $\boldsymbol{L}_{kj}$ is only piecewise smooth on Γ_k (partner elements change within a receiving facet), so an elevated sub-cell rule removes the consistency error of the default facet rule.
- **Consistent traction across the interface** (offset-surface construction): extract the consistent nodal reaction on the partner facet surface immediately exterior to Γ_k, where the one-sided partial assembly is exact, and transfer the weak traction to Γ_k's trace space with a surface-to-surface variational operator.
- **Single-operator characteristic transfer**: assemble the partner's complete Robin datum $-\tilde{\lambda} + \tilde{\boldsymbol{W}}(\boldsymbol{Z}\dot{\boldsymbol{u}}_j + \alpha \boldsymbol{u}_j)$ — where $\tilde{\lambda}$ is the weak consistent traction and $\tilde{\boldsymbol{W}}$ the boundary mass matrix, both on the offset surface — from the partner's own nodal values, and transfer it with one operator, so that any mesh-scale cancellation within the datum happens on the partner side, before transfer.

Table 4: Transfer-fidelity sequence on the nonconforming 1:0.75 pair at resolution h (1 ms, II, PU metric). Each increase in transfer fidelity increases the spurious interface work.

	dashpot work	net PU drift
pointwise + recovered stress	−607 J	−0.75%
variational + recovered stress	−726 J	+6.2% (growth)
variational, subdivided + recovered stress	−857 J	+2.4% (growth)
variational, subdivided + consistent traction	−1129 J	+9.3% (growth)
single-operator characteristic	−1084 J	+9.8% (growth)

The measured sequence (Table 4) shows that every fidelity increase enlarges the spurious interface work. The variational transfer does restore an h-dependence that pointwise interpolation lacks (dashpot work 726 → 484 J under refinement, versus unchanged), but the channel magnitudes remain of the same order, and at production resolution the net worsens: projecting the kinematics decorrelates the kinematic-transfer error from the recovered-traction error, breaking the cancellation that kept the pointwise scheme's net small. The characteristic-transfer result refutes the hypothesis that the operator pairing between traction and velocity is responsible for the residual. The conclusion is structural, not implementational: *on nonmatching grids, content that cannot be represented identically across the interface must be dissipated; smoothing the transferred data is beneficial, and nodal stress recovery had been providing exactly that stabilizing attenuation.*

5.2 Content-aware absorption: a conclusive null result

To turn that accidental stabilization into a designed mechanism, an additional dissipative self term $\boldsymbol{W}\,\boldsymbol{Z}\,(\boldsymbol{I} - \widehat{\Pi})\,\dot{\boldsymbol{u}}_k$ was implemented (`content aware absorption`), where $\widehat{\Pi}$ is the $\boldsymbol{W}$-orthogonal projection onto the span of the variational projector's columns — the transferable subspace at the boundary. The operator is verified: the filter $\boldsymbol{F} = \boldsymbol{I} - \widehat{\Pi}$ annihilates constants, vanishes identically on node-aligned interfaces and on the coarse side of the study pair, is nonzero only on the fine side ($\|\boldsymbol{F}\| = 0.44$), and is provably dissipative (asserted in test 106). **Its measured effect is null**: the absorber dissipates 4.6 J over 1 ms while the channels and the net remain identical to the baseline. Because the operator is verified, the null result is informative: *the recirculating content lies inside the transferable subspace.* Both grids can represent the recirculating content; the discrepancy arises because their discrete dispersion relations differ. No transfer-side construction can remove a dispersion discrepancy — which retroactively explains the entire fidelity sequence.

A final diagnostic confirms the mechanism: rerunning the pointwise baseline with a dissipative integrator (Newmark $\gamma = 0.6$, $\beta = 0.3025$) reduces the dashpot work from -607 to -231 J and the velocity jumps *decay* ($11.8 \to 5.6$ m/s between run halves) instead of growing ($33 \to 47$): the recirculation is suppressed by subdomain-level damping, not at the interface. (Newmark $\gamma = 0.6$ damps broadband: it also removed 87% of the physical energy on this benchmark.)

5.3 HHT-α damping and the associated energy loss

HHT-α numerical dissipation was added to the Newmark integrator (`HHT alpha:` $\bar{\alpha} \in [0, 1/3]$; internal force assembled at the blended state $(1-\bar{\alpha})\, u_{n+1} + \bar{\alpha}\, u_n$; $\gamma = 1/2 + \bar{\alpha}$ and $\beta = (1+\bar{\alpha})^2/4$ derived; $\rho_\infty = (1-\bar{\alpha})/(1+\bar{\alpha})$) and swept on both mesh pairs at 1 ms (Table 5).

Table 5: HHT-α sweep on the impedance overlap coupling (pointwise transfer, 1 ms). The interface-specific extra loss is the nonconforming loss beyond the damping-induced loss measured on the conforming pair, computed as $1 - (E/E_0)_{\mathrm{nonconf}}/(E/E_0)_{\mathrm{conf}}$.

$\bar{\alpha}$	0	0.02	0.05	0.1	0.2
nonconforming E/E_0	1.007	0.952	0.892	0.833	0.781
conforming E/E_0 (damping-induced loss)	0.999	0.969	0.938	0.905	0.873
interface-specific extra loss	±1%	1.7%	4.9%	8.0%	10.5%
dashpot work [J]	−607	−595	−580	−561	−540
jump growth (halves, m/s)	$33 \to 47$	$32 \to 44$	$31 \to 41$	$30 \to 37$	$29 \to 32$

Mild HHT only marginally reduces the interface channels even though $\bar{\alpha} = 0.1$ has the same $\rho_\infty = 0.818$ as the Newmark $\gamma = 0.6$ diagnostic: at $\Delta t = 10^{-6}$ s the mesh-frequency band sits at $\omega\Delta t \approx 0.3$–$0.7$, which the time step resolves and HHT (by design) preserves, while broadband first-order Newmark-γ damping reaches it, removing the physical content along with it (87% loss vs. 17% at the same ρ_∞). Enlarging the time step to $\Delta t = 4 \times 10^{-6}$ s, which moves the band into HHT's damped range, is ineffective on this benchmark: the damping-induced loss on the conforming pair grows to 31%, because the parabolic release has no scale separation — its physical content extends to the grid scale, so any damping strategy removes physical energy. In general, damping converts the non-robust ±1% net drift (sign-indefinite, growth-capable) into monotone, bounded decay, with an energy loss proportional to the fraction of the problem's energy in the recirculating band; production problems with smooth loading and resolution margin should obtain the robustness with small energy loss using $\bar{\alpha} \approx 0.02$–$0.05$.

6 Nonconforming overlap: the parametric energy study

6.1 Drift across integrators, overlap widths, and mesh ratios

Table 6 reports the drift $E(1\,\mathrm{ms})/E(0) - 1$ with pointwise transfer across integrator pairs, overlap widths, and mesh ratios.

Three regimes appear: (a) conforming implicit–implicit runs conserve to $\leq 0.1\%$ at every overlap width; (b) nonconforming runs lose tens of percent, worse with mesh ratio and with smaller overlap; (c) a minority of cases, including II at maximum overlap and 1:0.5, *gain* energy. Variational transfer improves the small-overlap losses substantially ($-44.5 \to -4.4\%$ at 0.0127/1:0.75) but regresses elsewhere ($+12.5 \to -50.7\%$ at 0.0508/1:0.5); it improved 17 of 24 nonconforming cases, reducing the mean |drift| from 33.4 to 25.4% — an improvement, but not a structural remedy.

Table 6: Energy drift at 1 ms of the impedance overlap coupling (pointwise transfer, PU metric), in percent.

overlap [m]	ratio	II	IE	EI	EE
0.0508	1:1.0	−0.07	−8.9	−0.2	+8.7
	1:0.75	+1.1	−8.7	−0.9	−12.0
	1:0.5	+12.5	−26.5	−53.8	−57.7
0.0254	1:1.0	−0.07	+5.0	−0.4	+10.2
	1:0.75	−28.4	−6.0	−22.3	−13.9
	1:0.5	−17.3	−18.5	−58.9	−45.7
0.0127	1:1.0	−0.02	−15.7	−6.1	+3.6
	1:0.75	−44.5	−40.6	−41.1	−31.4
	1:0.5	−63.8	−64.3	−70.8	−61.9

6.2 Channel-resolved interface work

Table 7 reports the interface work integrated over 1 ms for representative II runs, summed over both subdomains.

Table 7: Channel-resolved interface work of the impedance overlap coupling (1 ms, II runs, both subdomains). The conforming row uses the consistent traction (automatic on node-aligned interfaces); the nonconforming rows use recovered stress. Positive dashpot work in the fourth row proves the discrete dashpot term is not a quadratic form. The ΔE column is consistent with the drift values of Table 6; the third row's values, mistranscribed in a superseded working note, were confirmed by rerunning the instrumented case. The jump column is the run average of the per-stop RMS velocity jump defined in Sec. 4.

case	ΔE blended	dashpot work	traction work	jump RMS [m/s]
conforming (0.0254, 1:1.0)	−1 J	−1.8 J	+70 J	~ 1
nonconf. 0.0254, 1:0.75 (pointwise)	−428 J	−586 J	+92 J	~ 20
nonconf. 0.0127, 1:0.75 (pointwise)	−670 J	−1066 J	+347 J	~ 16
nonconf. 0.0508, 1:0.5 (pointwise)	+188 J	**+554** J	−211 J	~ 80
nonconf. 0.0127, 1:0.75 (variational)	−67 J	−864 J	+780 J	~ 26

Three facts follow. First, on conforming meshes the dashpot channel is negligible and the coupling is conservative: the velocity jump vanishes at Schwarz convergence. Second, on nonconforming meshes the dashpot channel carries the loss: the jump is an order of magnitude larger and never vanishes — the coarse trace space cannot represent the fine side's boundary velocity, and the dashpot persistently absorbs the difference, exactly as an absorbing boundary absorbs outgoing content. Third, and conclusive for identifying the mechanism: in the 0.0508/1:0.5 case the "dashpot" does +554 J of *positive* work. A dissipator can never do positive work; this proves the discrete dashpot term is not a quadratic form. Because each subdomain interpolates partner fields with its own, unrelated operator, the two sides' dashpot terms do not pair, and the total is sign-indefinite. The variational row explains why variational transfer improves only some of the cases: its dashpot still dissipates −864 J, but the traction channel returns +780 J — the improvement is an accidental near-cancellation of two large opposing channels, not a removal of either defect.

Scaling both impedances by $s \in \{1, 0.5, 0.25\}$ on the 0.0127/1:0.75 II case reduces the loss monotonically (−44.5, −40.5, −29.2%) but sublinearly: the dissipated power is $Z[\![\dot{u}]\!]^2$ and the jump itself grows as the absorber detunes, so the product falls more slowly than Z — the signature of an absorbing-boundary mechanism rather than of a fixed damping force.

6.3 Control experiment: Dirichlet matching

To separate a defect of the impedance condition from an error intrinsic to nonconforming overlap coupling, the same nine mesh configurations were run with Dirichlet overlap coupling (II), in both transfer variants (Table 8).

Table 8: Energy drift at 1 ms of Dirichlet overlap coupling (II). * run failed (Newton divergence) before 1 ms; drift at last stop.

overlap [m]	ratio	Dirichlet pointwise	Dirichlet weak
0.0508	1:1.0	+0.05%	—
	1:0.75	+5507%*	+2418%
	1:0.5	+2765%*	+5752%*
0.0254	1:1.0	−0.03%	—
	1:0.75	+423%	+749%
	1:0.5	+5628%*	+7765%*
0.0127	1:1.0	+0.01%	—
	1:0.75	+33%	+22%
	1:0.5	+3268%*	+262%

Hard kinematic matching conserves to within ±0.05% on conforming meshes and is severely unstable on every nonconforming one — and the weak (contractive) transfer does *not* stabilize it, consistent with the theory, since Dirichlet transmission violates the Robin requirement regardless of the transfer. Two consequences. First, the impedance dashpot's dissipation is not merely a defect: it is what stabilizes a coupling that hard kinematic exchange destabilizes; its magnitude is the energy in the inter-grid discrepancy it must continually absorb. Second, the Dirichlet instability grows with overlap width (opposite to the impedance losses), consistent with a feedback loop between the two interpolation surfaces through the overlap volume.

6.4 Mesh-ratio sweep: injection is generic

An independent parametric study observed significant energy injection at mesh ratios the original three-point family does not sample. The following sweep confirms and maps the effect: the clamped mesh is coarsened through $r = 1.0, 0.9, \ldots, 0.2$ at all three overlap widths and all four integrator pairs, with variational transfer throughout — 108 runs (Table 9). The 1:1.0 row reproduces the conforming baselines of Sec. 6.1, and the 1:0.5 row reproduces the earlier variational results at that ratio exactly.

Three observations refine the diagnosis.

Injection at mild coarsening is systematic, not exceptional. In the ratio band 1:0.9–1:0.7, the II column injects at nearly every overlap width (up to +29.8%), IE reaches +45.1%, and eight cells in the band lie at or above +7% — with the contractive variational transfer already in place. The contraction property bounds the transfer but not the sign of the dashpot work; no per-side transfer improvement yields sign-definiteness on the two-surface overlap coupling.

Table 9: Energy drift at 1 ms across the mesh-ratio family (impedance overlap coupling, variational transfer, PU metric), in percent.

	overlap 0.0508 m				overlap 0.0254 m				overlap 0.0127 m			
ratio	II	IE	EI	EE	II	IE	EI	EE	II	IE	EI	EE
1:1.0	−0.1	−8.9	−0.2	+8.7	−0.1	+5.0	−0.4	+10.2	−0.0	−15.7	−6.1	+3.6
1:0.9	+7.6	−11.7	−5.5	+8.5	−4.4	−8.1	+4.4	+9.1	+0.1	−14.8	−1.0	+4.3
1:0.8	**+29.8**	−15.3	+5.5	−9.5	+20.4	**+45.1**	−2.4	−1.9	−3.1	−20.3	−5.4	+4.7
1:0.7	+19.2	−12.8	+2.7	−11.6	+11.8	−18.6	+0.7	−5.4	+11.1	+7.3	−9.8	−17.9
1:0.6	−61.1	−54.0	−57.7	−40.4	+2.1	−18.6	−63.9	−53.4	−4.5	−34.2	−57.1	−52.7
1:0.5	−50.6	−32.3	−36.4	−56.8	−11.0	−17.6	−55.7	−54.3	−29.2	−54.0	−56.8	−52.0
1:0.4	−12.4	+3.9	−43.1	−53.1	−52.7	−50.2	−57.0	−63.8	−36.6	−52.4	−78.2	−76.2
1:0.3	−65.0	−44.5	−78.3	−57.4	**+173.7**	**+274.7**	−81.2	−78.8	−30.5	−58.6	−79.5	−79.4
1:0.2	−79.1	−76.8	−80.8	−78.9	−66.5	−66.2	−86.3	−87.4	−22.2	−45.8	−91.2	−90.3

A strong injection case. At 0.0254/1:0.3 the coupling drives the beam to +174% (II) and +275% (IE) of $E(0)$, growing smoothly for ∼0.7 ms and saturating near 3–4 × $E(0)$ — steady energy injection at the interface rather than a numerical divergence. At this ratio the clamped mesh has degenerated to a single-element cross-section, but degeneracy alone does not decide the outcome: the same clamped mesh at quarter overlap dissipates −30.5%, and the explicit-clamped pairs on the identical configuration dissipate −81%. Whether the sign-indefinite interface work injects or absorbs is a property of the mesh pair and the overlap geometry together.

The drift is strongly non-monotone in the ratio. At 0.0254 overlap, II moves −52.7 → +173.7 → −66.5% across the three adjacent ratios 1:0.4 → 1:0.3 → 1:0.2. There is no usable regime map of the form "mild contrast implies small drift": both the sign and the magnitude of the energy defect are highly sensitive to the discrete mesh pair.

This resolves the question of whether the energy error of the nonconforming overlap coupling can be bounded a priori: it cannot. The interface work is sign-indefinite in practice as well as in principle, and both signs are realized at large amplitude across neighboring points of a smooth parameter family.

7 Requirements for energy consistency

Four independent lines of analysis agree on the structure of the problem and of its solution.

7.1 The interface-work pairing identity

For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure [4, Eq. (34)]; Karimi–Nakshatrala [6, Eq. (84)])

$$E_{\mathrm{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} [\![\dot{\boldsymbol{u}}_k]\!]^T \boldsymbol{C}_k^T [\![\Lambda_k]\!], \tag{13}$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid; in the notation of those references, Δt_k is the side-k time step, $\boldsymbol{C}_k$ the interface restriction (constraint) matrix of side k, Λ_k the interface traction impulse transmitted to side k, and $[\![\cdot]\!]$ the change of the bracketed quantity across the coupling step. The term vanishes if and only if the traction is a single-valued field acting equal-and-opposite on the two sides through *adjoint* trace operators, with velocity continuity enforced at instants shared by both time grids. Gravouil–Combescure prove their subcycled coupling makes $E_{\mathrm{int}} \le 0$ (dissipative, never a source); Prakash–Hjelmstad [5] restructure the intermediate tractions so it telescopes to exactly zero. Neither treats spatially nonconforming interfaces; both warn that breaking the pairing leaves E_{int} of indefinite sign — precisely the +554 J observation of Sec. 6.2.

7.2 Representability: the converged nonconforming solution

Gander, Halpern, and Nataf [7] analyze discrete Schwarz waveform relaxation with impedance transmission on nonmatching grids. Their key statements: the choice $Z = \rho c$ of the partner side is the exact transparent operator in one dimension and optimal-in-class generally, so the impedance choice is not the problem; their nonmatching-grid convergence proof requires the intergrid transfer to be an L^2 contraction (a projection qualifies; pointwise interpolation does not); and, for nonconforming grids, the coupled system *defines* the solution when converged — there is no monodomain scheme it equals. Continuity at the converged fixed point holds only through the projection: the component of the fine trace orthogonal to the coarse trace space is not representable by the partner and is never matched. The impedance dashpot then acts on that persistent component as a genuine absorbing boundary — sign-definite dissipation that is required for stability, not an iteration defect. The remedy they and Achdou, Japhet, Maday, and Nataf [8] point to is not a better Z but a *common trace space* for the datum entering the dashpot.

7.3 The telescoping condition and the adjoint requirement

Energy-stable discontinuous-Galerkin couplings [9, 10, 11] make the interface energy identity exact by construction: a single Riemann state is computed on a common description of the interface (the mesh intersection) and enters both sides' weak forms; the interface power then telescopes to

$$P \;=\; -\int_\Gamma [\![\dot{\boldsymbol{u}}]\!]^T \boldsymbol{Z} [\![\dot{\boldsymbol{u}}]\!] \,\mathrm{d}S \;\le\; 0 \qquad \text{(or exactly 0 for the central variant)}, \tag{14}$$

with the dissipation confined to the jump and vanishing under refinement and at convergence. The precise discrete requirement: if side k receives partner kinematics through $\Pi_{j\to k}$, the transfers must be L^2-adjoints through the variational cross-mass matrix,

$$\boldsymbol{W}_1 \Pi_{2\to 1} \;=\; \big(\boldsymbol{W}_2\, \Pi_{1\to 2}\big)^T \;=\; \boldsymbol{B}, \qquad B_{mn} = \int_\Gamma \varphi^1_m\, \varphi^2_n \,\mathrm{d}S, \tag{15}$$

with $\boldsymbol{W}_k$ the side-k boundary mass matrices of Sec. 1 and the integrals evaluated on the common refinement. Two independent pointwise interpolations violate (15), the cross terms do not cancel, and the interface term has no sign. Liu et al. [11] additionally warn that nonconforming cross-interpolation matrices are severely ill-conditioned and that interface unknowns may need implicit treatment even between explicit interiors (their IMEX-Newmark) — directly relevant to the conforming explicit-member drifts of Table 6 (EE gains +3.6 to +10.2%; IE is sign-indefinite, −15.7 to +5.0%), where the explicitly evaluated dashpot at lagged time levels injects energy that central difference ($\gamma = \frac{1}{2}$, zero algorithmic damping) never removes.

7.4 Synthesis: why each observed regime occurs

Conforming, implicit. The transfer is the identity, the adjoint condition (15) holds trivially, the jump vanishes at Schwarz convergence, the dashpot does no work: conserved.

Nonconforming, dissipative majority. The jump cannot vanish (representability); the dashpot absorbs the unrepresentable content persistently. The loss grows with mesh ratio and shrinks with overlap width (a wider overlap separates the boundary from fine-scale content and gives the coarse subdomain more elements to express the transferred field).

Nonconforming, growth cases. Non-adjoint pairing makes the dashpot term sign-indefinite; wherever the positive branch dominates, the coupling injects instead of absorbing. The ratio sweep shows this is generic and that the sign changes between adjacent mesh ratios: structural, not stochastic.

Conforming, explicit free subdomain. A time-level, not spatial, pairing failure: the dashpot force is evaluated explicitly at the lagged partner velocity, and central difference has no algorithmic damping to absorb the error.

Variational improvement. The variational projection is a contraction and the characteristic transfer restores much of the traction consistency, but the dashpot still acts on the full jump; the net outcome depends on the near-cancellation of two large channels.

8 Design conclusions for the overlap family

The theory and the measurements of Secs. 5 and 6 yield four conclusions for the overlap coupling.

Sign-definiteness is structurally unavailable. The adjoint condition (15) presumes a *shared* interface on which the two sides' works telescope. The overlap variant has none: data are exchanged on two distinct surfaces Γ_1, Γ_2, so no pairing of their works exists, and energy consistency could come only from agreement of the two discrete solutions in the overlap — exactly what the dispersion discrepancy of Sec. 5.2 precludes at fixed h. No transfer-side construction can make the overlap interface work sign-definite, and a universal dashpot sign test is not an attainable regression criterion for this variant.

The contractive transfer is the implementable requirement. `transfer: variational` is therefore the default for the impedance overlap condition (pointwise interpolation remains an explicit legacy option; the two coincide on node-aligned interfaces). Empirically (Secs. 6.1 and 6.2) the variational transfer converts the worst sign-indefinite case (+554 J of positive dashpot work) into dissipation of the correct sign and improves 17 of 24 nonconforming cases, but it cannot eliminate the dispersion-jump dissipation, and isolated growth cells persist.

Restricting the dashpot to the representable jump is ineffective. The redesign also tested replacing $\boldsymbol{Z}[\![\dot{\boldsymbol{u}}]\!]$ by $\boldsymbol{Z}\,\widehat{\Pi}\,[\![\dot{\boldsymbol{u}}]\!]$, where $\widehat{\Pi}$ is the W-orthogonal projection onto the transferable span (`representable dashpot: true`; a strict no-op on node-aligned interfaces). The premise was that the persistent jump is trace content the coarse space cannot represent, so that filtering it would let it reflect (conserving) instead of being absorbed. Measurement falsifies the premise: on the 0.0127/1:0.75 case the fine side's $\widehat{\Pi}$ is a genuine projection ($\|\widehat{\Pi} - I\|/\sqrt{n} = 0.48$) yet the dashpot work is unchanged ($-1066 \to -1065$ J) — the jump lies *inside* the representable subspace, consistent with the dispersion diagnosis: the two discrete solutions differ by a smooth, in-band field that no static spatial filter can remove. Where $\widehat{\Pi}$ does act (0.0508/1:0.5), per-side filtering worsens the transfer asymmetry and the drift ($+12.5 \to +45.6\%$). The option is retained (default off) as a diagnostic of where the jump lies.

The remaining theoretical ingredients belong to the nonoverlap variant. The adjoint pairing itself, the harmonic-mean (Riemann-weighted) pair impedance, and the implicit treatment of interface terms for explicit subdomains all require a single shared interface; their implementation is the subject of Sec. 9.

The channel-resolved interface-work instrumentation remains the regression metric for the overlap family: the dashpot work must be ≈ 0 on conforming meshes at Schwarz tolerance, and the drift tables of Sec. 6 are the acceptance benchmark for any future change to the transmission machinery — a small net drift can be a cancellation of large opposing channels.

9 The adjoint-paired nonoverlap coupling

The theory identifies the nonoverlap impedance variant — one shared interface Γ, transmission condition (6) — as the member of the family on which the adjoint condition (15) is available. It has been implemented and measured on the cantilever benchmark, split at $x = 0.127\,\mathrm{m}$, with the mesh-ratio family of Sec. 6.4.

9.1 Baseline: the legacy coupling

The pre-existing nonoverlap impedance coupling (per-side transfer operators, per-side impedance and Robin parameter) loses 2–18% at mild mesh contrast across the four integrator combinations (Fig. 1) — *including on conforming meshes* (−9.6% II at 1:1) — and injects strongly at strong contrast (+223% EI and +363% EE at 1:0.3; +77% II at 1:0.2). Probes isolate the conforming loss: it is unchanged under $\alpha \to 0$, under equalized α, and under $\Delta t \to \Delta t/2$, with the Schwarz iteration converged to 10^{-8}. It is therefore a *consistency* defect of the fixed point itself: the coupling transfers the partner's static reaction $-\boldsymbol{f}^{\mathrm{int}}$, but in dynamics the two sides' static reactions are not equal and opposite — they differ by the interface nodes' inertia, each side carrying its own tributary mass. The coupled fixed point then differs from the monodomain discretization by a term of the order of the interface tributary mass times the interface acceleration, measured as a steady, Δt-independent interface energy loss.

9.2 The paired coupling

Enabling `adjoint pairing: true` (the default for `Schwarz impedance nonoverlap`; `adjoint pairing: false` restores the legacy transfer) changes three things:

1. the exchanged traction becomes the dynamically consistent d'Alembert reaction $\boldsymbol{M}\,\boldsymbol{a} + \boldsymbol{f}^{\mathrm{int}} - \boldsymbol{f}^{\mathrm{body}}$, so the two interface rows sum exactly to the monodomain row and the monodomain trajectory is the fixed point;
2. both sides derive their transfer operators from *one* shared cross-mass matrix B (integrated over the finer side's facets), with $\Pi_1 = W_1^{-1}B$, $\Pi_2 = W_2^{-1}B^T$, and each side's force transfer the adjoint of the partner's kinematic transfer ($N_1 = \Pi_2^T$, $N_2 = \Pi_1^T$), which is exactly (15);
3. the pair shares one impedance — the harmonic mean $2Z_1Z_2/(Z_1+Z_2)$ of the per-side material impedances $Z_k = \rho_k c_{p,k}$ (each multiplied by its side's `impedance scale` before the mean), the Riemann-solver weighting of [10, 11] — and one Robin parameter α.

On parameter selection: the impedance is *estimated* from material data (Sec. 2.2), the Robin parameter is *prescribed* in the input (`robin parameter`, default 0 under `Schwarz impedance nonoverlap`; the classical `Schwarz RR nonoverlap` keyword instead requires it positive, Sec. 2.1), and no estimate of α is attempted because none is needed for consistency. At the Schwarz fixed point the two Robin conditions add to traction balance and subtract to $Z\,\dot{\boldsymbol{\delta}} + \alpha\,\boldsymbol{\delta} = \mathbf{0}$ for the interface jump $\boldsymbol{\delta}$; with $\boldsymbol{\delta}(0) = 0$ the converged solution is independent of $Z > 0$ and $\alpha \geq 0$ — both act only on the convergence rate of the iteration and on the interface energy bookkeeping during it. The pair impedance is scalar (P-wave); a tensor P/S pair impedance analogous to (11) has not been needed on this homogeneous benchmark and belongs with the bimaterial study of open question (iv). What consistency does require is that α be one number per interface: the Robin spring stores and returns energy conservatively only when both sides' interface forces pair through the same coefficient. Mismatched values under pairing are therefore rejected as an input error — Norma aborts, naming both side sets and values — since silently averaging them would apply a condition neither side stated.

Discrete exchange and assembly. For reproduction in another code, the complete discrete construction is as follows. The cross-mass matrix $\boldsymbol{B}$ of (15) is integrated over the facets of the *finer* side (identified as the side with more interface nodes): each facet quadrature point of the finer side is mapped to the partner surface by closest-point projection onto the partner's side set in the reference configuration, and the partner's facet shape functions are evaluated at the projected point. This finer-side facet rule approximates the common-refinement integral of (15): it is exact when the coarser trace mesh is nested in the finer facets, and carries facet-quadrature error wherever a coarse basis function kinks inside a fine facet. The partition-of-unity errors $\max_i |(\Pi_k \mathbf{1} - \mathbf{1})_i|$ are computed and logged at setup as a quadrature certificate (zero on conforming and nested interfaces); the non-nested failure window of Sec. 11 is the regime where this certificate is worst. The kinematic transfers are the dense solves $\Pi_1 = \boldsymbol{W}_1^{-1}\boldsymbol{B}$ and $\Pi_2 = \boldsymbol{W}_2^{-1}\boldsymbol{B}^T$, the force transfers their partner adjoints $N_1 = \Pi_2^T$ and $N_2 = \Pi_1^T$, and all transfers are scalar, applied componentwise to the three Cartesian components.

The weak partner reaction of side j is

$$\lambda_j \;=\; -\big(\boldsymbol{M}_j\,\ddot{\boldsymbol{u}}_j + \boldsymbol{f}_j^{\mathrm{int}} - \boldsymbol{f}_j^{\mathrm{body}}\big)\big|_{\Gamma}, \tag{16}$$

the restriction to side j's interface degrees of freedom of its assembled global d'Alembert residual, with $\boldsymbol{M}_j$ the mass matrix side j's integrator uses (consistent for Newmark, lumped for central difference). Side k's boundary force receives the partner datum

$$\boldsymbol{g}_k \;=\; N_k\,\lambda_j \;+\; Z\,\boldsymbol{W}_k\,\Pi_k\,\dot{\boldsymbol{u}}_j \;+\; \alpha\,\boldsymbol{W}_k\,\Pi_k\,\boldsymbol{u}_j \;=\; N_k\,\lambda_j \;+\; Z\,\boldsymbol{B}_k\,\dot{\boldsymbol{u}}_j \;+\; \alpha\,\boldsymbol{B}_k\,\boldsymbol{u}_j, \tag{17}$$

where $\boldsymbol{B}_1 = \boldsymbol{B}$ and $\boldsymbol{B}_2 = \boldsymbol{B}^T$ (the second form uses $\boldsymbol{W}_k\Pi_k = \boldsymbol{B}_k$), and all partner fields are evaluated at side k's current substep time through the temporal interpolation of Sec. 10 (in a same-step run the interpolation is trivial at the shared stop end). The self terms are not part of this datum: the tangent contribution $(Zc_v + \alpha)\,\boldsymbol{W}_k$ and the self force $\boldsymbol{W}_k(Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k)$ are assembled together with the subdomain's internal force at every residual evaluation, so the Newton (or explicit-update) residual sees the transmission condition exactly. When the Schwarz iteration relaxes this exchange, the relaxed iterate is the assembled datum $\boldsymbol{g}_k$ itself, on one side of the pair only (the later-listed subdomain of the pair); a fresh relaxation slot starts from zero rather than seeding with the incoming datum, which is why this path has no degenerate first-sweep fixed point.

Table 10 and Fig. 1 compare the two couplings. Three properties, in order of importance. First, *no injection at any ratio*: the interface work is sign-definite, as the telescoping identity (14) guarantees. Second, near-conservation through mild contrast (-0.03% at 1:0.9, where every other variant tested loses or gains several percent). Third, the residual dissipation at strong contrast is the genuine trace-space jump being absorbed at rate $Z\int_\Gamma [\![\dot{\boldsymbol{u}}]\!]^2$: at the exactly nested ratio 1:0.5,

Table 10: Energy drift at 1 ms of the nonoverlap impedance coupling, implicit–implicit, legacy vs. adjoint-paired, in percent.

ratio	legacy II	paired II
1:1.0	−9.61	**−0.03**
1:0.9	−6.84	**−0.03**
1:0.8	−7.96	−3.59
1:0.7	−2.06	−6.55
1:0.6	−9.39	−9.71
1:0.5	−13.25	−13.32
1:0.4	−45.78	−15.13
1:0.3	+18.95	−15.08
1:0.2	+77.49	−13.26

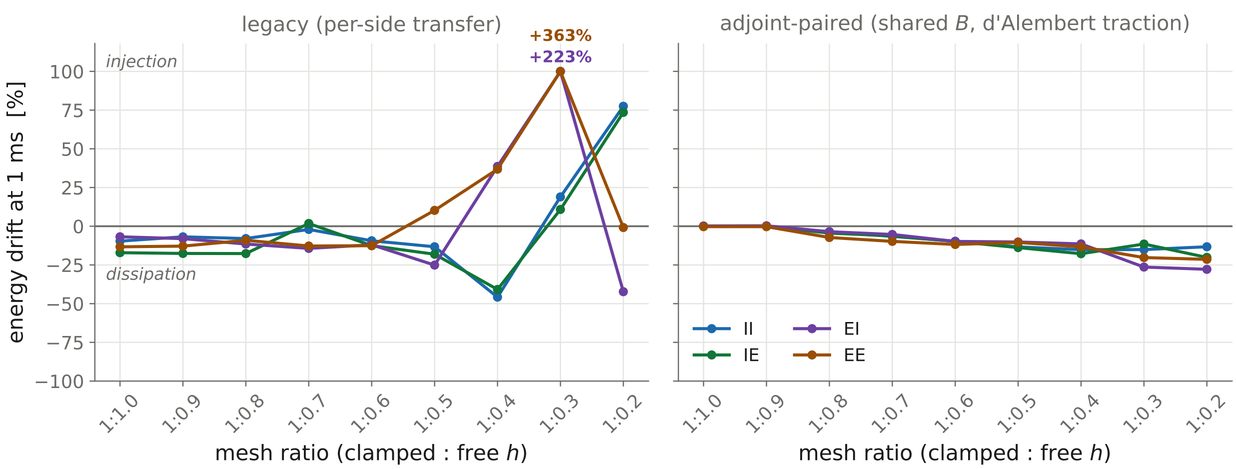

Figure 1: Nonoverlap impedance Schwarz, energy drift at 1 ms across the mesh-ratio family, all four integrator combinations. Left: legacy per-side transfer (consistent-mass integer-step metric) — sign-indefinite, with injection up to +363% (EE) and +223% (EI) at 1:0.3. Right: the adjoint-paired coupling (staggered lumped-mass metric of Eq. (3) for explicit members) — monotone, sign-definite trace-jump dissipation at every ratio, and conservation to $\leq 0.3\%$ on conforming and 1:0.9 meshes.

where the legacy transfer is coincidentally adjoint and the quadrature exact, paired and legacy agree to 0.1% — the remaining loss is not a transfer artifact but the representability-induced loss of Sec. 7.2, now realized in a controlled, monotone, sign-definite form. (Why the legacy coupling's static-reaction loss does not additionally appear at this ratio has not been isolated; the agreement is reported as measured.)

9.3 Explicit subdomains: IMEX interface treatment

With an explicit (central-difference) subdomain on either side, the paired coupling is strongly unstable ($+10^3$ to $+10^5$ %) if the interface terms are evaluated at lagged time levels: the consistent traction couples the partner's *acceleration* into the interface force, an algebraic loop that the implicit pairs resolve within the Schwarz iteration. The remedy is the implicit–explicit (IMEX) structure of Liu et al. [11]: because the dashpot force is linear in the unknown acceleration, the explicit update is closed implicitly on the interface rows only,

$$\big(\boldsymbol{M} + \gamma\,\Delta t\,Z\,W\big)\big|_\Gamma\,\boldsymbol{a}_{n+1} \;=\; \boldsymbol{M}\,\boldsymbol{a}^{\mathrm{expl}}\big|_\Gamma, \tag{18}$$

a small symmetric-positive-definite solve per interface (reused for the three components) while the interior stays matrix-free explicit (`imex_interface_acceleration!`); the dashpot then acts on the end-of-step velocity and the Schwarz iteration synchronizes both sides at t_{n+1} exactly as in the implicit–implicit case. In (18), $\boldsymbol{M}\big|_\Gamma$ is the diagonal (lumped) mass restricted to the interface rows, $\boldsymbol{W}$ the side's own boundary mass matrix, γ the central-difference velocity-update coefficient ($\frac{1}{2}$), and $\boldsymbol{a}^{\mathrm{expl}}$ the plain explicit acceleration update, whose right-hand side already contains the dashpot evaluated at the predictor velocity $\tilde{\boldsymbol{v}}$ and the Robin spring at the (known) end-of-step displacement — the interface solve replaces the predictor-velocity dashpot by the end-of-step one on the interface rows only. Interface degrees of freedom held by Dirichlet conditions keep their prescribed acceleration and enter the free rows' right-hand side as data.

Table 11: Energy drift at 1 ms of the adjoint-paired nonoverlap coupling, all integrator combinations, staggered metric (3) for explicit members, in percent.

ratio	II	IE	EI	EE
1:1.0	−0.03	−0.11	+0.20	−0.27
1:0.9	−0.03	−0.10	+0.29	−0.27
1:0.8	−3.6	−4.4	−3.5	−7.3
1:0.7	−6.6	−6.2	−5.3	−9.8
1:0.6	−9.7	−9.8	−9.8	−11.8
1:0.5	−13.3	−13.8	−10.2	−10.5
1:0.4	−15.1	−17.8	−11.4	−13.3
1:0.3	−15.1	−11.5	−26.4	−20.2
1:0.2	−13.3	−20.1	−27.8	−21.4

On conforming and mildly nonconforming meshes all four integrator combinations conserve to $\leq 0.3\%$ (Table 11); the nonconforming columns carry the same monotone, sign-definite trace-jump dissipation in every combination. (The small positive EI entries, +0.2–0.3%, are bounded transient storage in the conservative interface spring, not secular growth.) In the consistent-mass integer-step metric the explicit columns instead read −4 to −37%; the difference is the metric artifact of Sec. 1.2, not the coupling — against the monodomain explicit reference the conforming paired EE trajectory is exact to 0.35% *pointwise over the whole millisecond*: the coupled trajectory is the monodomain trajectory, as the fixed-point argument predicts.

10 Subcycling: different time steps per subdomain

Different time steps per subdomain require no new exchange machinery: each subdomain records its per-substep trajectory during every Schwarz iteration, the coupling evaluates the partner's state in time at each of its own substeps, and the paired d'Alembert traction follows. The recorded trajectory consists of full-field snapshots of displacement, velocity, acceleration, and assembled internal force at every substep, anchored by a snapshot of the stop-start state pushed before the first substep (without the anchor, a non-subcycling partner's history is a single end-of-stop entry and the exchange degenerates to constant, end-sampled data). When a coupling condition is evaluated at time t, each recorded field is interpolated piecewise-linearly to t (queries at or beyond the history endpoints clamp to the endpoint values); the partner model's state is temporarily replaced by the interpolants, the coupling terms are evaluated — so the d'Alembert reaction (16) is formed from the *interpolated* acceleration and internal force with the constant mass matrix, not by interpolating a stored reaction — and the partner state is then restored. (Enabling this exposed a progressive time-step reduction defect in the subcycler — the stop-aligned remainder step persisted as the nominal step and the Schwarz re-integration amplified the floating-point remainder geometrically — fixed independently of the pairing.)

10.1 Genuine interpolation

The exchange was intended to interpolate the partner piecewise-linearly in time — the interpolated-continuity structure under which Gravouil–Combescure [4] prove the subcycled interface work dissipative ($E_{\text{int}} \leq 0$, $O(\Delta t)$) — but two infrastructure defects meant that is not what actually ran. First, the history interpolation selected the wrong segment (an off-by-one error that returned the end-of-history value for any query in the last segment and extrapolated backward elsewhere). Second, the per-stop history was anchored at the first substep, not the stop start, so a non-subcycling partner's history was a single end-of-stop entry. Net effect: both directions of the exchange end-sampled the partner's stop-end state — constant, not interpolated, data. With both defects fixed the exchange is genuinely piecewise-linear in both directions. Table 12 compares the two at 1 ms with the free side subcycled 4:1 ($\Delta t = 0.25\,\mu$s inside a 1 μs stop; consistent-mass integer-step metric for comparability across the development history).

Table 12: Energy drift at 1 ms of the subcycled (4:1) adjoint-paired coupling with the defective end-sampled exchange vs. the corrected piecewise-linear interpolation, in percent (consistent-mass integer-step metric).

case	end-sampled (defective)	interpolated (fixed)
conforming II	−2.8	−3.6
conforming IE	−9.6	−10.1
conforming EE	−10.6	−9.6
1:0.5 II	−20.6	**−14.1**
1:0.5 IE	−24.4	**−21.3**

(In the staggered metric the interpolated exchange reads conforming II −3.6, IE −1.4, EE −1.2%: the additional energy loss of explicit subcycling is about one percent on conforming meshes.)

Genuine interpolation is distinctly more accurate at mesh contrast, where end-sampling aliases the fine side's intra-window motion into the exchanged traction. It performs slightly worse on the conforming implicit cases because the end-sampled datum, being the stop-end value applied throughout the window, *leads* the true trajectory, and that anticipatory bias is a mild energy-injection channel that coincidentally offset part of the interpolation dissipation. A cancellation between two discretization errors is not a conservation property, and this one fails precisely at strong mesh contrast; the interpolated exchange is therefore the default behavior of the paired coupling.

10.2 Exact telescoping is incompatible with a partitioned exchange

Prakash–Hjelmstad [5] telescope the subcycled interface work to exactly zero by giving the interface traction a linear-in-time representation whose step values satisfy $\frac{1}{2}(\hat{\lambda}_n + \hat{\lambda}_{n+1}) = \overline{\lambda}_{n\to n+1}$ (the partner's window average), so the trapezoidal work sums of the two sides cancel window by window. Three Schwarz-compatible constructions of this datum were implemented and measured:

Window average. Supplying the coarse side with the partner's trapezoidal window average is work-conjugate to the fine side's interpolation — but an average estimates the window midpoint, and the half-window lag of the interface force behind the interface motion is artificial damping: uniformly worse than plain interpolation ($-3.6 \to -4.2$, $-10.1 \to -11.1$, $-9.6 \to -12.3$, $-14.1 \to -15.5$, $-21.3 \to -23.0$).

Exact telescoping recursion. The lag-free datum $\hat{\lambda}_{n+1} = 2\overline{\lambda} - \hat{\lambda}_n$ satisfies the telescoping identity exactly — and diverges: the factor of two on the partner-state channel doubles the Schwarz iteration's contraction factor, and the coupled iteration diverges within tens of stops.

Least-squares endpoint fit. The endpoint value of the least-squares linear fit of the partner trajectory over the window is lag-free, filters intra-window content instead of aliasing it, and telescopes for window-linear tractions. It is stable over tens of stops and passes the short regression test — and over hundreds of stops it injects energy without bound ($+10^3$–10^5% before the run aborts): a zero-lag filtered estimate is an extrapolation, and its frequency response exceeds unity outside the linear band.

The pattern is the classical stability principle of partitioned coupling: only *causal* transfer operators (lag ≥ 0) are robust in a fixed-point exchange, and every zero-lag work-conjugate datum is anticipatory. Prakash–Hjelmstad's telescoping is consistent with this — their method solves the coupled interface problem *directly* (a FETI-style interface system per large step), which is precisely what a partitioned Schwarz sweep does not do. Exact subcycled interface-work telescoping therefore requires a direct interface solve, not a modified partitioned exchange; within the Schwarz family, genuinely interpolated Gravouil–Combescure transfer is the measured stable optimum, and its $O(\Delta t)$ dissipation (-3.6% at conforming II vs. -0.03% same-step) is the energy loss intrinsic to subcycling. (The window-average and endpoint-fit utilities remain in `interpolation.jl` as tested building blocks.)

11 Long-horizon behavior

Every conclusion above rests on 1 ms runs — about two transits of the release front. To test which behaviors are steady and which are transients, all 151 study configurations were rerun to 10 ms (10^4 stops): the full overlap sweep, the nonoverlap paired sweep and its subcycled set, and fresh monodomain references. Both monodomain references conserve the staggered energy (3) to $\pm 0.000\%$ over the full 10 ms, and every surviving run reproduces its recorded 1 ms drift at its $t = 1$ ms row.

11.1 Overlap: the 1 ms drifts were transients of instabilities

The overlap family is long-horizon unstable almost everywhere. Of the 108 overlap runs, 74 were terminated by element inversion after exponential energy growth (typically +3,000 to +10,000% of E_0 at failure, between 2 and 9 ms), and most of the remainder decayed toward zero energy. The sign-indefinite drifts of the 1 ms sweep were early sections of these trajectories: configurations that appeared dissipative at 1 ms (e.g. maximum overlap 1:0.6 II, −61%) reversed and grew to +5,432% at failure. Mesh conformity does not stabilize the mixed-integrator combinations: the conforming half-overlap IE run read +13% at 1 ms, grew exponentially, and failed at 6.25 ms; conforming EI behaves alike at all three widths. The only overlap configurations that remain stable and accurate at 10 ms are the conforming II runs: +0.64, +0.61, +0.16% at the three overlap widths (Fig. 2). The overlap coupling is therefore a short-horizon method on nonconforming meshes, and for non-II combinations even on conforming ones.

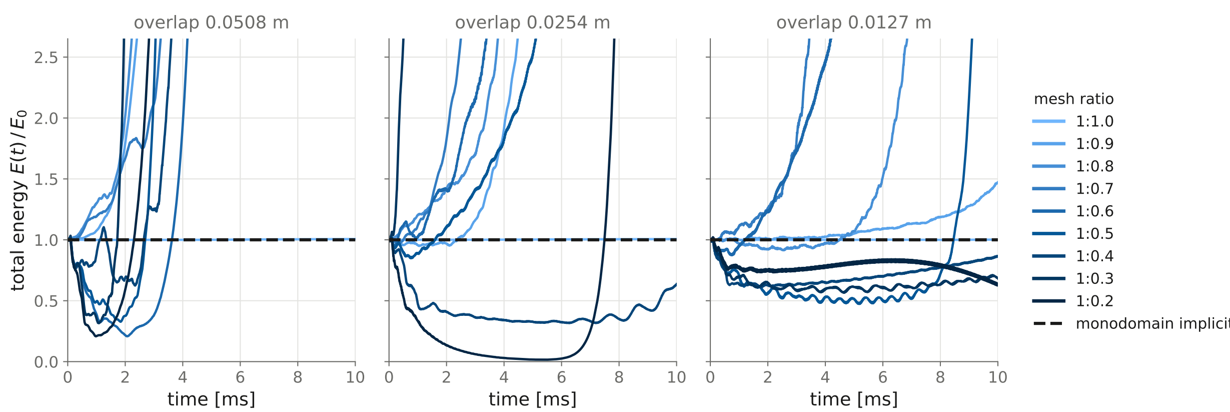

Figure 2: Overlap impedance Schwarz (variational transfer), implicit–implicit, 10 ms horizon. Nearly every nonconforming configuration either grows exponentially until element inversion ends the run (curves leaving the top of the axes) or decays toward zero energy; only the conforming (1:1.0) runs track the monodomain reference. The mixed and all-explicit combinations (not shown) are worse: they lose even the conforming cases.

11.2 Nonoverlap pairing: no injection, but accumulating dissipation and two new failure modes

The paired nonoverlap coupling never injects at any ratio in any combination, at 10 ms as at 1 ms. Its monotone trace-jump dissipation, however, accumulates (Table 13).

Three observations (Fig. 3). First, with an implicit member the conforming and 1:0.9 columns remain within 0.00 to −0.9% over ten thousand stops. Second, at strong contrast ($\leq$ 1:0.6) the sign-definite dissipation that read as 10–28% at 1 ms reaches 40–85% by 10 ms: bounded and ordered, but far too large for wave-transit accuracy. Absorption of unrepresentable content proceeds at a persistent rate; it is not a fixed initial loss. Third, the mildly nonconforming, *non-nested* ratios 1:0.7 and 1:0.8 form a failure window not detectable at 1 ms: the paired transfer accumulates interface-

Table 13: Energy drift at 10 ms of the adjoint-paired nonoverlap coupling, staggered metric, in percent. † t: run terminated by element inversion at t ms. $*$: run ended in the noise-saturated explicit–explicit state discussed in the text.

ratio	II	IE	EI	EE
1:1.0	+0.00	−0.9	−0.2	$*$
1:0.9	+0.00	−0.5	+0.1	$*$
1:0.8	† 5.2	† 3.9	† 4.0	† 2.6
1:0.7	† 4.9	† 3.7	† 3.9	† 2.8
1:0.6	−47.3	−56.4	−42.9	$*$
1:0.5	−56.8	−72.0	−40.8	$*$
1:0.4	−54.9	−73.9	−40.4	$*$
1:0.3	−65.1	−66.0	−81.1	$*$
1:0.2	−53.3	−73.1	−84.1	$*$

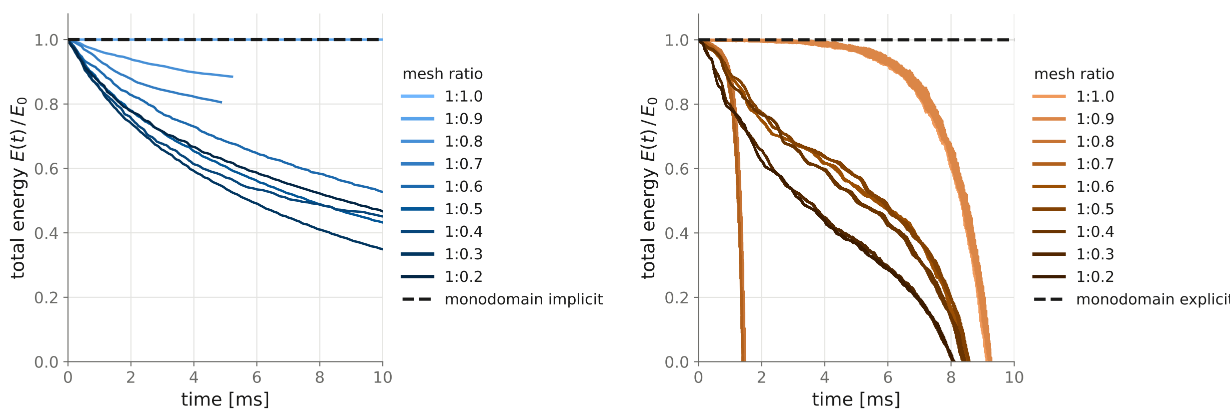

Figure 3: Adjoint-paired nonoverlap impedance Schwarz, 10 ms horizon, staggered energy metric. Left: implicit–implicit — no injection anywhere; conforming and 1:0.9 track the monodomain reference, strong contrast dissipates monotonically, and the non-nested ratios 1:0.7–1:0.8 end early at ≈ 5 ms (interface element inversion). Right: explicit–explicit — the staggered kinetic term goes secularly negative at every ratio as undamped grid-frequency content accumulates; the 1:0.7–1:0.8 runs fail by central-difference instability at 2.6–2.8 ms.

local high-frequency displacement content, and every integrator combination eventually fails there — explicit members at 2.6–4 ms by central-difference instability (despite the 3.7× stability margin of Sec. 1.2, so the instability is driven by the accumulated content, not by the time step), implicit pairs at $\approx$ 5 ms when the first interface element inverts while the global energy is still smooth (−19% at failure). This is not a relaxation artifact: reruns with fixed $\theta = 0.5$ fail identically (trajectories matching to three digits), so the noise accumulation is attributable to the transfer operators themselves. The nested ratios (1:0.5 exactly, and the strong ratios, whose transfer is strongly smoothing) do not exhibit it.

11.3 Explicit–explicit saturation, and subcycling as a remedy

The EE column degrades at every ratio by 10 ms, conforming included, ending with the staggered metric near $-2\,E_0$. The end state is not a divergence: the stored energy stays bounded and physical throughout (the conforming run maintains $E/E_0 \geq 0.97$ to 5 ms, then declines), but the staggered kinetic term goes steadily negative — the signature of undamped high-frequency content accumulating in the velocity and acceleration fields. This is not an artifact of operating near the stability limit: the explicit subdomains run at 0.27 of theirs (Sec. 1.2). Central difference with $\gamma = 1/2$ has no algorithmic dissipation at any stable step, so whatever grid-frequency content the interface exchange introduces is never removed; with no implicit member in the pair, it accumulates without bound in amplitude-neutral modes. The measured remedy is subcycling the explicit member: the subcycled conforming EE run (free side at $\Delta t/4$) ends at −10.8% with no saturation, consistent with the mild dissipation of the interpolated subcycled exchange (Sec. 10.1), which removes the high-frequency content that the non-dissipative same-step exchange retains. Subcycling has the opposite effect on the implicit–implicit pair: the same-step conforming II run conserves to +0.00% while the subcycled one loses −20.2% (at 1:0.5, −61.6%) — the Gravouil–Combescure interpolation dissipation of Sec. 10.1, harmless per stop, accumulates over 10^4 stops. At long horizons the multi-time-step exchange is thus a trade-off: it suppresses the explicit saturation and degrades implicit pairs through interpolation dissipation. Figure 4 shows both effects.

The long-horizon ranking is unambiguous: adjoint-paired nonoverlap coupling with at least one implicit interface member, on conforming or nested mesh ratios, is the only tested configuration that is stable and accurate at 10 ms; the overlap family is short-horizon only.

12 Windowed controller stops and the relaxation state

The multi-domain controller advances in *stops*; within a stop each subdomain subcycles to the stop end with its own time step, and partner data are exchanged as per-substep trajectories with piecewise-linear time interpolation (Sec. 10). With the subdomain steps held fixed (1 μs), does the controller stop size — the window over which the Schwarz iteration converges before committing — affect the converged physics? A sweep over windows of 1 (baseline), 2, 5, and 10 μs on representative overlap and nonoverlap cases, run to 10 ms, gave a split answer.

Overlap: a pure cost parameter. Every overlap trajectory is identical across all four windows to all printed digits — the conforming II run ends at +0.61% in all four, and the 1:0.5 instability fails at the same instant in all four. The overlap exchange interpolates partner kinematics from history with no per-pair relaxation state, so the windowed waveform iteration converges to the same-step fixed point.

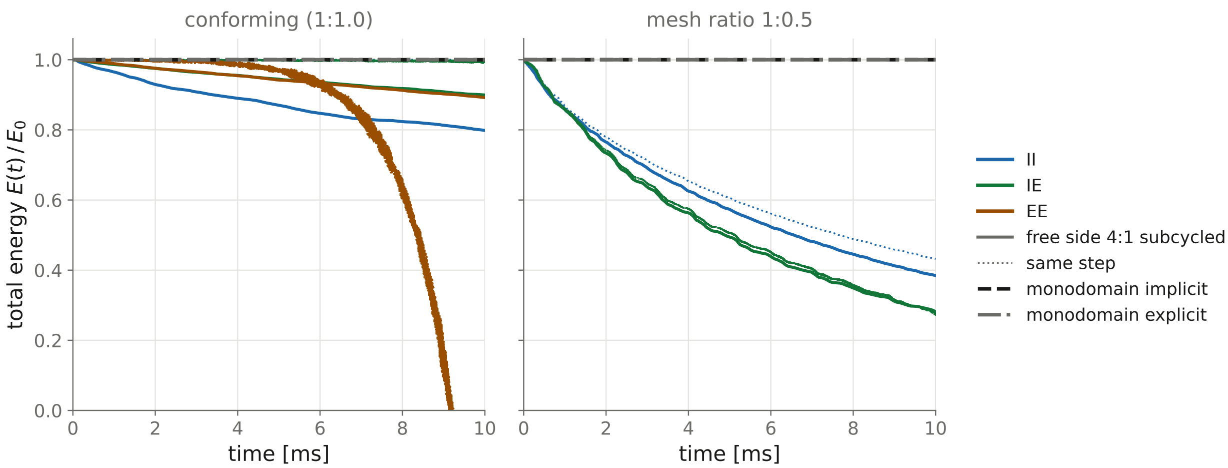

Figure 4: Subcycled (free side 4:1, solid) vs. same-step (dotted) adjoint-paired nonoverlap coupling over 10 ms. Left, conforming: the subcycled EE run (orange solid, $\Delta t/4$) avoids the saturation that terminates same-step EE (orange dotted), while subcycled II (blue solid) incurs the accumulating interpolation dissipation that same-step II (on the reference line) does not. Right, mesh ratio 1:0.5: at strong contrast the trace-jump dissipation dominates and the additional subcycling dissipation is comparatively small (II subcycled -61.6% vs. same-step -56.8%).

Nonoverlap: the window shifted the fixed point. Drift for the conforming paired II case at 10 ms: $+0.00\%$ same-step, $+11.3\%$ at the $2\,\mu$s window, -81.9% at $5\,\mu$s, -83.7% at $10\,\mu$s (Aitken relaxation; with fixed $\theta = 0.5$ the sequence is a monotone energy loss instead, -7.1% at $2\,\mu$s to -87.9% at $10\,\mu$s). This was not a convergence failure — the sweeps converged normally — but a defect of the relaxation machinery: the relaxation state (the stored boundary-force vectors and the Aitken residual buffers) was kept as one vector per interface pair, updated on every boundary-condition application. In a windowed stop the relaxed side of a pair (the later-listed member) applies its coupling condition once per *substep*, so the blend $(1-\theta)\,g + \theta\,\mathrm{rhs}$ mixed iterates across *time* rather than across Schwarz sweeps: a causal exponential low-pass filter (lag $\sim \Delta t_{\mathrm{sub}}/\theta$) on the exchanged traction, whose lag dissipation grows with the window. Under Aitken the defect changes sign: the θ estimate compares residuals belonging to different times, overshoots past 1, and the resulting phase lead injects energy — the $+11.3\%$ entry. (The subcycled study of Sec. 10 was unaffected because its subcycling free side is listed first and therefore unrelaxed.)

Two experiments isolate and confirm the mechanism. First, $\theta = 1$ — which uses no relaxation state at all — reproduces the same-step physics on the unmodified code at every window ($+0.00\%$ over the full 10 ms at the $10\,\mu$s window). Second, with the fix below in place, all windowed policies converge to the same-step trajectory. The fix keys the relaxation state by pair *and* by substep time slot within the current stop, clearing it at every stop (`relaxation_slot!` in `schwarz.jl`); with one substep per stop it reduces exactly to the classical per-pair iteration, and the same-step code path is bit-identical to the pre-fix baselines. Windowed runs then match same-step to Schwarz truncation (the residual 3.9×10^{-6} displacement mismatch at working tolerances falls to 3.7×10^{-8} as the stopping tolerances tighten), and a windowed-equivalence regression guards this in the adjoint-pairing test (test 110).

Relaxation policy inside windows. With the state fixed, the remaining question is which relaxation policy a windowed stop should use. Measured on the conforming II case at the $10\,\mu$s

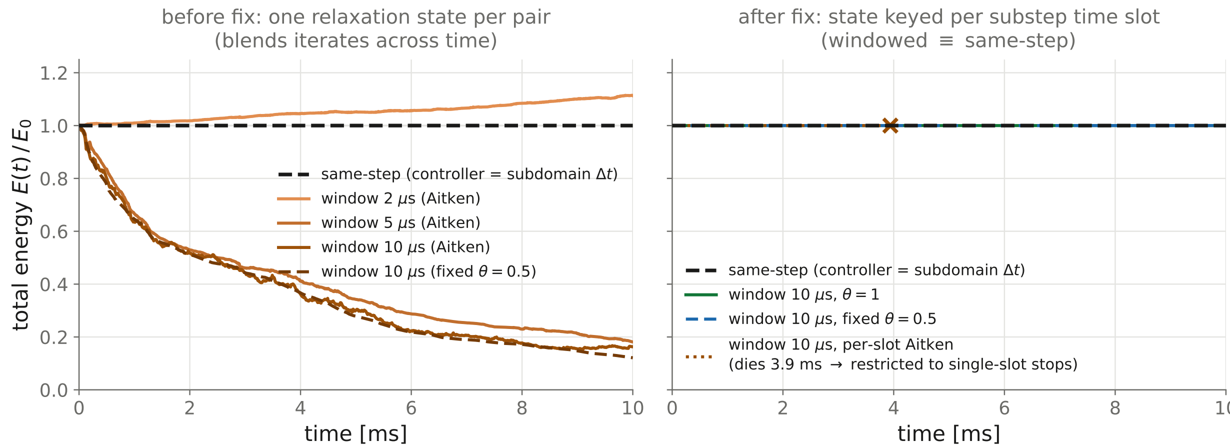

Figure 5: Windowed controller stops on the conforming paired II case, 10 ms. Left, pre-fix: the per-pair relaxation state blends iterates across time — Aitken injects at the 2 μs window and dissipates at 5–10 μs; fixed $\theta = 0.5$ dissipates monotonically. Right, post-fix (slot-keyed state): every windowed policy reproduces the same-step trajectory; the per-slot Aitken run is physically correct until its unclamped θ excursions cause a Newton failure at 3.9 ms ($\times$), which motivated restricting Aitken to single-slot stops.

window (mean Schwarz sweeps per stop): $\theta = 1$ converges in 19.1 and runs the full 10 ms without failure; per-slot Aitken takes $\sim$25 and fails at 3.9 ms when an unclamped θ excursion causes the Newton iteration to diverge; fixed $\theta = 0.5$ takes 47.5 (no failures); and a waveform-pooled Aitken (one θ from residuals pooled across the window's slots) takes 55.1 — no failures on II but it fails on EE, where it holds θ persistently negative. Aitken acceleration therefore offers no benefit inside windows and can cause failure, and Norma applies it only to single-slot stops, where its $\approx 3\times$ sweep reduction is established (`aitken_applies`); windowed stops use the configured `relaxation parameter`, with $\theta = 1$ recommended — the impedance dashpot supplies the contraction, so no under-relaxation is needed.

Two limitations remain. The all-explicit pair at large windows is impractical under any policy: the windowed explicit sweep map is nearly non-contractive (fixed $\theta = 0.5$ reaches the 128-sweep limit and still fails at 9.8 ms). And windows are never a cost optimization for the nonoverlap exchange: the number of substep solves per unit time is (sweeps per stop)/Δt_{sub} regardless of the window, and the windowed sweep counts above all exceed the same-step Aitken count, so the controller step is purely a convenience parameter (e.g. for aligning output stops), cost-neutral at best. For the overlap family, by contrast, larger stops are a mild speedup (fewer, marginally more iterated stops). One pre-fix measurement is explicitly withdrawn as an artifact: windowed EE with fixed θ at the 2 μs window appeared nearly conservative (-0.25% at 10 ms), but this was the defect's low-pass filter accidentally damping the grid-frequency saturation, not a property of the coupling; post-fix, windowed EE saturates exactly like same-step EE, and the effective remedy remains subcycling the explicit member (Sec. 11).

Open issue: a θ-sensitive fixed point. The windowed-equivalence test exposed a pre-existing defect, orthogonal to the fix: the *converged* paired-impedance solution depends weakly on the relaxation parameter — $\theta = 0.5$ and $\theta = 1.0$ differ by $\sim 4 \times 10^{-5}$ relative displacement on the nonconforming (1:0.5) cantilever at a 10^{-13} stopping tolerance, same-step and windowed alike, and an Aitken run coincides with the $\theta = 1$ solution to 10^{-12}. Instrumentation rules out the blend algebra (the blended base is identically zero at interface degrees of freedom, because `apply_bcs` zeroes the boundary force before each pass); the evidence points to a neutral — non-contracting — mode of the paired exchange that retains its θ-dependent initialization and is not detected by the displacement-based stopping criterion. The magnitude is far below every signal in this study ($\geq 10^{-4}$), but it is a genuine correctness question and warrants a separate investigation.

13 Conclusions and recommendations

13.1 Summary of findings

1. On a shared interface, Robin coupling is fully reflecting ($|R| \equiv 1$) and the impedance condition is absorbing ($|R| < 1$, zero at $\alpha = 0$) with the impedance fixed by material data; in the quasi-static limit the impedance condition reduces to Robin (Sec. 2).

2. On an overlap boundary, the impedance condition is consistent with the monodomain solution only with the partner-traction term, and the quality of that traction sets the conforming energy error: $59\% \rightarrow 14\% \rightarrow 1.5\% \rightarrow 0.08\%$ over 1 ms as the traction evaluation improves to the consistent nodal reaction, with no adjustable parameters (Sec. 4).

3. On nonconforming overlap decompositions the coupling family is absorbing-or-unstable: Dirichlet matching (pointwise or weak) is severely unstable, and the impedance condition is stable precisely because its dashpot continually absorbs the inter-grid discrepancy, whose dominant component is coarse-grid dispersion — a smooth, in-band solution defect that neither spatial filtering nor any measured increase in transfer fidelity can eliminate at fixed h (Secs. 5, 6).

4. The interface work of the overlap coupling is sign-indefinite in principle (two independent, non-adjoint transfers) and in practice (injection up to +275%; sign changes between adjacent mesh ratios), and at a 10 ms horizon 74 of 108 nonconforming overlap runs fail. The energy error of the nonconforming overlap coupling admits no a priori bound in sign or magnitude (Secs. 6.4, 11).

5. Energy consistency requires a single-valued interface traction acting through L^2-adjoint transfer operators built from one cross-mass matrix — then the interface power telescopes to a sign-definite dissipation on the interface jump (14). This is implementable only on the nonoverlap variant, which has a shared interface (Sec. 7).

6. The adjoint-paired nonoverlap coupling — d'Alembert interface reaction, shared cross-mass matrix, harmonic-mean pair impedance, IMEX interface rows for explicit subdomains — never injects at any mesh ratio in any integrator combination, conserves to $\leq 0.3\%$ at 1 ms on conforming and mildly nonconforming meshes in all four combinations, and reproduces the monodomain trajectory pointwise (Sec. 9).

7. At long horizons the paired coupling's sign-definite dissipation at strong mesh contrast accumulates (40–85% at 10 ms); the non-nested ratios 1:0.7–1:0.8 accumulate interface-local high-frequency content until an interface element inverts; and the all-explicit pair saturates with undamped grid-frequency content, for which subcycling the explicit member is an effective remedy (Sec. 11).

8. Subcycling uses genuine piecewise-linear interpolation of partner trajectories (two silent defects that reduced it to end-sampling are fixed); its $O(\Delta t)$ dissipation is the energy loss

intrinsic to subcycling, and exact interface-work telescoping is structurally incompatible with a partitioned exchange — it requires a direct interface solve (Sec. 10).

9. The Schwarz relaxation state must be keyed per pair and per substep time slot; otherwise controller stops larger than a subdomain step blend iterates across time and shift the converged fixed point. With the slot-keyed state, windowed stops are equivalent to same-step to Schwarz truncation, and the controller step is a pure convenience parameter. Aitken acceleration applies only to single-slot stops; windowed stops should use $\theta = 1$ (Sec. 12).

13.2 Practical recommendations

- **Energy-critical dynamic coupling**: use the adjoint-paired nonoverlap impedance variant (the default for `Schwarz impedance nonoverlap`) with at least one implicit interface member and a conforming or nested mesh ratio. Avoid the non-nested mild ratios (1:0.7–1:0.8 class) for long runs.
- **Overlap decompositions**: conforming meshes with implicit–implicit integration conserve energy at any horizon; every other overlap configuration is suitable for short transients only. On conforming node-aligned interfaces the consistent traction makes the impedance overlap condition exact to the Dirichlet-coupling reference.
- **All-explicit interfaces**: subcycle the finer side; central difference with $\gamma = 1/2$ does not remove the grid-frequency content the exchange introduces at any stable step, and only the dissipative subcycled exchange suppresses its accumulation.
- **Parameters**: no impedance tuning anywhere — the impedance is material data (harmonic mean per pair for nonoverlap; partner material with P/S split for overlap). The Robin parameter affects only the iteration, not the converged solution, and must be a single value per interface under pairing. Historical per-problem tuning of impedance scales and large Robin values is obsolete.
- **Diagnostics**: use the blended (partition-of-unity) energy output and the staggered metric (3) for explicit members; use the channel-resolved interface-work instrumentation as the acceptance metric for any change to the transmission machinery — a small net drift can be a cancellation of large opposing channels.
- **Robustness at fixed cost** on nonconforming overlap runs that cannot be avoided: mild HHT-α damping ($\bar{\alpha} \approx 0.02$–$0.05$) converts sign-indefinite drift into monotone decay, with an energy loss proportional to the energy in the affected frequency band.

13.3 Open questions

(i) The weakly θ-sensitive fixed point of the paired exchange ($\sim 4 \times 10^{-5}$ relative displacement), evidence of a neutral mode not detected by the displacement-based stopping criterion (Sec. 12). (ii) The non-nested mesh-ratio failure window of the paired transfer at long horizons (Sec. 11). (iii) Exact subcycled telescoping via a direct FETI-style interface solve, together with the localized-Lagrange-multiplier frame construction [12, 13] (Sec. 10.2). (iv) A bimaterial benchmark to exercise the tensor impedance and the harmonic-mean pair impedance where they should be most beneficial. (v) A smooth-content benchmark to quantify the HHT-α robustness versus energy-loss trade-off on problems with scale separation (Sec. 5.3).

Data provenance

Correction-sequence study and instrumentation: `rigel:~/test/Norma/cantilever-energy`. Parametric study, ratio sweep, 10 ms horizon, controller-step sweep, figures, and run harness: `~/cantilever-energy-study/` (`ratio-sweep/`, `*-10ms`, `controller-step-sweep/`, `wccm-plots/`, `drift-table-10ms.txt`). Regression tests (indices as of 2026-08): 104 (overlap impedance energy and consistent-traction patch, `schwarz-overlap-dynamic-cantilever-impedance-energy.jl`), 105 (tensor impedance cross-scaling), 106 (variational transfer projectors and content-absorption dissipativity), 110 (adjoint pairing, windowed equivalence, and the keyword guards, `test/schwarz-nonoverlap-impedance-adjoint-pairing.jl`).

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